MICROKERNEL ARCHITECTURE MANUAL

HYDRA OS

UNIX-like Distributed Microkernel for Modern UEFI Hardware

BOOT SEQUENCE PIPELINE

UEFI POST → GDT/TSS → CPU Diagnostics → Frame Allocator → PML4 Paging → Capability Space → IPC Rendezvous

Edition: v0.1.0 Architecture Specification

Target Architecture: x86_64-unknown-uefi (Native 64-bit Long Mode)

Security Model: Capability-Based Access Control (seL4-style)

IPC Subsystem: Synchronous & Asynchronous Message Rendezvous

Generated: October 25, 2026

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SYSTEM OVERVIEW • BOOT SEQUENCE ORDER #01

Hydra OS - Codebase Documentation Index

Welcome to the technical documentation for **Hydra OS**. Every source code file in the repository is documented with real-world analogies, conceptual explanations, architectural diagrams, and API breakdowns designed to be accessible to anyone with general programming knowledge.

🗺️ Documentation Directory Map

```
docs/
├── README.md                    # This Index
├── DOCUMENTATION_STANDARD.md    # Rustdoc-compatible source code documentation standard
├── build.md                     # Build automation script (build.rs)
└── src/
    ├── main.md                  # Kernel entry point & bootstrap orchestrator (src/main.rs)
    ├── console.md               # Thread-safe console & formatting macros (src/console.rs)
    ├── vfs.md                   # Unix File Descriptors, Pipes, Syscalls & Pipelines (src/fs/)
    ├── zfs.md                   # Microkernel ZFS File Server & SPA CoW (src/fs/zfs/)
    ├── arch/
    │   ├── mod.md               # Architecture-specific module root (src/arch/mod.rs)
    │   ├── boot_services.md     # Hardware takeover & ExitBootServices (src/arch/boot_services.rs)
    │   ├── gdt.md               # Global Descriptor Table, TSS & Rings (src/arch/gdt.rs)
    │   ├── idt.md               # Interrupt Descriptor Table & Exception Gates (src/arch/idt.rs)
    │   ├── cpu.md               # Control registers & CPU diagnostics (src/arch/cpu.rs)
    │   └── x86_64/
    │       └── boot.md          # Legacy BIOS boot assembly (header.asm, main.asm, main64.asm)
    ├── memory/
    │   ├── mod.md               # Memory management module root (src/memory/mod.rs)
    │   ├── frame_allocator.md   # Physical memory manager & 4 KiB frames (src/memory/frame_allocato
    │   ├── heap.md              # Dynamic kernel heap allocator & alloc crate (src/memory/heap.rs)
    │   └── paging.md            # Virtual memory & 4-level PML4 translation (src/memory/paging.rs)
    ├── capability/
    │   ├── mod.md               # Capability types & Rights bitflags (src/capability/mod.rs)
    │   └── table.md             # Task capability table & CSpace (src/capability/table.rs)
    ├── video.md                 # Hardware GOP display driver & framebuffer console (src/video/)
    ├── shell.md                 # Interactive microkernel shell & command REPL (src/shell.rs)
    └── ipc/
        ├── mod.md               # IPC subsystem root (src/ipc/mod.rs)
        ├── message.md           # Fast register payload & transferred caps (src/ipc/message.rs)
        └── endpoint.md          # Message rendezvous channels (src/ipc/endpoint.rs)
```

📚 Module Reference Table

Subsystem	Source File	Documentation	Key Concepts & Analogies
Boot & Main	`src/main.rs`	docs/src/main.md	Entry point, boot sequence, hardware diagnostics, and panic handler.
Console I/O	`src/console.rs`	docs/src/console.md	Thread-safe `print!` / `println!` macros, keyboard input, screen formatting.
Video & GOP	`src/video/`	docs/src/video.md	UEFI GOP linear framebuffer, 8x16 bitmap font, and standalone graphical console.
Interactive Shell	`src/shell.rs`	docs/src/shell.md	Command REPL, real-time memory/CPU inspection, CSpace manipulation, IPC testing.
GDT & TSS	`src/arch/gdt.rs`	docs/src/arch/gdt.md	Ring 0 / Ring 3 privilege levels, Task State Segment, double-fault stack.
IDT & Traps	`src/arch/idt.rs`	docs/src/arch/idt.md	256-gate IDT, assembly exception trampolines, IST double fault isolation, #BP/#PF dispatchers.
Hardware Takeover	`src/arch/boot_services.rs`	docs/src/arch/boot_services.md	ExitBootServices, GDT/TSS switch, autonomous CR3 PML4, and PS/2 keyboard takeover.
CPU Control	`src/arch/cpu.rs`	docs/src/arch/cpu.md	Control registers (`CR0` , `CR2` , `CR3` , `CR4`), paging roots.
Arch Root	`src/arch/mod.rs`	docs/src/arch/mod.md	Architecture abstraction layer for hardware independence.

Subsystem	Source File	Documentation	Key Concepts & Analogies
Physical Memory	`src/memory/frame_allocator.rs`	docs/src/memory/frame_allocator.md	UEFI memory map parsing, 4 KiB frame allocator ("parking lot attendant").
Dynamic Heap	`src/memory/heap.rs`	docs/src/memory/heap.md	Dynamic kernel heap allocator, `#[global_allocator]`, `Box` / `Vec` / `String` support.
Virtual Memory	`src/memory/paging.rs`	docs/src/memory/paging.md	4-level x86_64 paging tree (PML4), address translation, page protection.
Memory Root	`src/memory/mod.rs`	docs/src/memory/mod.md	Physical and virtual memory coordinator.
Capability Core	`src/capability/mod.rs`	docs/src/capability/mod.md	Security keycards (`Rights::READ`, `WRITE`, `EXECUTE`, `GRANT`, `CALL`).
Capability Table	`src/capability/table.rs`	docs/src/capability/table.md	Per-task CSpace keyring, permission checking, access revocation.
IPC Message	`src/ipc/message.rs`	docs/src/ipc/message.md	Registered courier package: 4 fast inline register arguments + security token.
IPC Endpoint	`src/ipc/endpoint.rs`	docs/src/ipc/endpoint.md	Postal mailbox rendezvous point, FIFO circular ring buffer.
IPC Root	`src/ipc/mod.rs`	docs/src/ipc/mod.md	Inter-process communication module export.
Build Automation	`build.rs`	docs/build.md	NASM assembly build script and link configuration.
Legacy Boot	`src/arch/x86_64/boot/`	docs/src/arch/x86_64/boot.md	Multiboot2 header and 32-bit to 64-bit assembly trampoline.

🚀 Boot Sequence Chronological Topic Order

When understanding or reviewing Hydra OS from initial hardware power-on through microkernel readiness, the documentation is structured sequentially:

1. **Firmware Handoff & Bootstrapping**: - Legacy Boot Trampoline & Long Mode Switch (Multiboot2 header, 32-bit to 64-bit Long Mode switch) - Build Automation & Assembly Pipeline (NASM toolchain and UEFI link settings)
2. **UEFI Entry & Bootstrap Orchestration**: - Kernel Entry Point & Bootstrap Pipeline (UEFI `main()`, panic handler, subsystem coordination, shell handover)
3. **Early Diagnostic Output & Keyboard Input**: - UEFI Console & Formatting (`print!`, `println!`, thread-safe `CONSOLE` spinlock, keyboard polling, cursor management)
4. **CPU Architecture & Hardware Isolation (Step 1 & 2)**: - Architecture Abstraction Layer (Hardware abstraction root) - Global Descriptor Table (GDT) & TSS (Ring 0 / Ring 3 privilege levels, Task State Segment, Double Fault IST stack) - Interrupt Descriptor Table (IDT) & Hardware Exceptions (256-gate IDT, assembly trampolines, IST double-fault stack, #BP/#PF traps) - Hardware Takeover & ExitBootServices (Termination of UEFI firmware, GDT/TSS switch, CR3 PML4 takeover, PS/2 keyboard) - CPU Control Registers & Diagnostics (CR0 protection, CR2 fault addresses, CR3 PML4 paging root, CR4 extensions)
5. **Physical & Virtual Memory Management (Step 3 & 4)**: - Memory Subsystem Coordinator (Memory management root) - Physical Memory & Frame Allocator (UEFI memory map parsing, 1 MiB low memory reservation, 4 KiB frame allocator) - Virtual Memory & 4-Level Paging (x86_64 PML4 translation, 2 MiB / 1 GiB huge page decoding)
6. **Capability Security Model (Step 5)**: - Capability Security Primitives (Security keycards, `Rights` bitflags, `CapType` kernel resource identifiers) - Task Capability Space & Table (Per-task CSpace table, fixed 256-slot array storage, rights verification)
7. **Inter-Process Communication (Step 6)**: - IPC Subsystem Root (Microkernel message passing architecture) - IPC Fast Register Messages (4 fast inline register payload words, labels, transferred capability tokens) - IPC Endpoints & Mailbox Rendezvous (Circular FIFO ring buffers, rendezvous channels, thread-safe `SharedEndpoint`)
8. **Interactive User Space & Subsystem Diagnostics**: - Interactive Microkernel Shell (Command REPL, real-time memory/CPU inspection, CSpace manipulation, IPC testing, RTC clock, reboot/shutdown)

📄 Generating the Unified Documentation PDF

You can generate a single, professional PDF manual containing all documentation files, automatically formatted and ordered by the boot sequence topics:

Windows:

```
generate_pdf.bat
```

or

```
python scripts\generate_docs_pdf.py
```

Linux / WSL / macOS:

```
make docs-pdf
```

or

```
python3 scripts/generate_docs_pdf.py
```

The compiled PDF is generated at `docs/Hydra_OS_Documentation.pdf` and mirrored to `dist/Hydra_OS_Documentation.pdf`. An HTML version suitable for browser inspection or custom printing is also created at `docs/Hydra_OS_Documentation.html`.

SYSTEM OVERVIEW • BOOT SEQUENCE ORDER #02

Hydra OS — Source Code Documentation Standard

- **Document Version**: 1.0.0
- **Scope**: All Rust source files (`src/**/*.rs`) across the entire Hydra OS codebase.
- **Engine Compatibility**: `rustdoc` (Rust 2021 Edition), Markdown (CommonMark), GitHub Flavored Markdown (GFM).
- **Primary Goal**: Ensure every module, type, function, and unsafe block is maximally understandable, self-explanatory, and accessible to both systems engineers and learners.

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1. Core Philosophy: "Code Tells How, Docs Tell Why"

1. **Self-Explanatory Architecture**: Source code expresses the raw algorithmic logic; documentation articulates the **architectural intent**, **hardware constraints**, **invariants**, and **safety prerequisites**.
2. **Accessible Rigor**: Use clear, accessible English accompanied by intuitive systems analogies (e.g., capability tokens as security keycards, frame allocator as parking lot attendant), while maintaining absolute technical precision regarding bit masks, memory layouts, and register states.
3. **Rustdoc First**: All public items must compile cleanly into formatted HTML via `cargo doc` without broken links, unescaped HTML, or missing sections.
4. **No Trivial Comments**: Comments like `// sets the value` or `/// returns the id` are forbidden. Every doc comment must provide genuine insight beyond what the type signature already conveys.

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2. Doc Comment Types & Scope

Syntax	Target	Required Scope
`//!`	**Module / Crate Root**	Top of every file: subsystem role, architectural dependencies, dataflow.
`///`	**Types, Functions, Constants**	All `struct` , `enum` , `trait` , `fn` , `const` , `static` , and type aliases.
`//`	**Implementation Logic**	Complex internal algorithms, hardware errata workarounds, assembly rationale.

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3. Module-Level Documentation Standard (`//!`)

Every single `.rs` file **must begin** with an outer module-level documentation block (`//!`) structured as follows:

```
//! # Subsystem Name: High-Level Title
//!
//! Concise summary of the module's core responsibility in 1–2 sentences.
//!
//! ## Architectural Role & Design Rationale
//! Explanation of why this module exists, how it fits into the microkernel design,
//! and the problems it solves. Include real-world analogies where helpful.
//!
//! ## Key Types & Interfaces
//! - [`TypeName`](TypeName): Summary of primary structure.
//! - [`function_name()`]: Core public entry point.
//!
//! ## Hardware & Dependency Context
//! - CPU Privilege Level: Ring 0 (Kernel) / Ring 3 (Userland)
//! - Hardware Interaction: I/O Ports, MSRs, MMIO, or Page Tables
//! - Concurrency: Lock-free atomics / `spin::Mutex` / Interrupt-safe
//!
//! ## Dataflow & Architectural Diagram (Optional but Strongly Encouraged)
//! ```text
//! +--------------+         +--------------+         +--------------+
//! | Input Stream | ------> | Pipe Buffer | ------> | File Output |
//! +--------------+         +--------------+         +--------------+
//! ```
```

---

4. Function & Method Documentation Standard

Every function (both `pub` and non-trivial `pub(crate)`/private functions) must adhere to the **Standard Section Pattern**:

```
/// Single-sentence summary written in the third-person imperative mood.
///
/// Multi-paragraph detailed explanation of the function's internal mechanism,
/// edge-case handling, and architectural side effects (e.g., changes to task state,
/// hardware interrupt flags, or memory page mappings).
///
/// # Arguments
///
/// * `param1` - Purpose, units (bytes, ticks, ms), valid range, and constraints.
/// * `param2` - Expected invariant or state requirement.
///
/// # Returns
///
/// Description of the return value. For `Option<T>` or `Result<T, E>`, explain
/// what `Some` / `None` or `Ok` / `Err` specifically signify in this context.
///
/// # Errors
///
/// Explicitly list all error conditions, negative errno values, and recovery steps:
/// - `-EBADF`: The provided file descriptor index was out of bounds or unopened.
/// - `-EMFILE`: The per-process file descriptor table is completely saturated.
///
/// # Panics
///
/// Explicitly state conditions that cause a panic, or guarantee:
/// - *Guaranteed to never panic under normal execution.*
///
/// # Safety
///
/// **MANDATORY FOR ALL `unsafe fn`!** State caller invariants:
/// 1. Pointer validity: Alignment, dereference bounds, and lifetime requirements.
/// 2. Interrupt context: Must interrupts be disabled (`CLI`) before invocation?
/// 3. Hardware state: Required CPU mode (e.g., 64-bit Long Mode, CR0.WP active).
///
/// # Examples
///
/// ```rust,no_run
/// let (read_fd, write_fd) = sys_pipe().expect("Failed to create pipe");
/// sys_write(write_fd, b"Hydra OS").unwrap();
/// ```
```

---

5. Struct, Enum, and Field Documentation Standard

Structs

Every struct must explain its **lifecycle**, **ownership**, and **threading guarantees**:

```rust
/// Circular ring buffer implementing an anonymous unidirectional streaming pipe.
///
/// # Concurrency & Thread Safety
/// This structure is protected by a [`spin::Mutex`] and is safe to share
/// across threads via an [`Arc<Mutex<PipeBuffer>>`]. Read and write operations
/// acquire the lock briefly to copy bytes into or out of the internal array.
///
/// # Lifetime & Cleanup
/// When all reader file descriptors drop (`readers == 0`), subsequent write calls
/// return a Broken Pipe error (`-EPIPE`). When all writers drop (`writers == 0`),
/// subsequent reads return `0` bytes (EOF) once the internal buffer is drained.
pub struct PipeBuffer {
    /// Internal 4096-byte ring storage for unread stream payload.
    buffer: [u8; PIPE_BUFFER_SIZE],
    /// Ring index where next incoming byte will be written (head pointer).
    head: usize,
    /// Ring index where next available byte will be read (tail pointer).
    tail: usize,
    /// Total count of valid, unconsumed bytes currently residing in the ring.
    count: usize,
    /// Number of active reader file descriptors holding a reference to this pipe.
    readers: usize,
    /// Number of active writer file descriptors holding a reference to this pipe.
    writers: usize,
}
```

Enums

Every enum and each individual variant must be documented:

```rust
/// Execution privilege level of an architectural thread/task.
#[derive(Debug, Clone, Copy, PartialEq, Eq)]
pub enum PrivilegeMode {
    /// Ring 0: Full hardware privilege with supervisor segment selectors (CS=0x08, SS=0x10).
    Kernel,
    /// Ring 3: Unprivileged userland execution with user segment selectors (CS=0x23, SS=0x1B).
    /// All hardware I/O and privileged instructions cause a General Protection Fault (#GP).
    User,
}
```

---

6. Rustdoc Formatting Rules & Intra-Doc Links

1. **Intra-Doc Links**: Always link to types using Rustdoc bracket syntax rather than raw markdown URLs: - `[Task](crate::task::Task)` - `[PipeBuffer](crate::fs::pipe::PipeBuffer)` - `[SCHEDULER](crate::task::scheduler::SCHEDULER)`
2. **Code Blocks**: Always provide a language specifier on triple-backtick blocks: - `rust,no_run` for Rust examples - `text` for ASCII diagrams - `bash` for shell commands
3. **Section Headings**: Use Markdown Level 1 (`# Section`) inside doc comments for standard sections (`# Arguments` , `# Returns` , `# Errors` , `# Safety`).

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7. Concrete Examples: Good vs. Bad Documentation

Example 1: Function Documentation

❌ Bad Documentation (Lazy / Inadequate)

```rust
// duplicates a descriptor
pub fn dup2(&mut self, oldfd: usize, newfd: usize) -> Result<usize, &'static str> {
    // ...
}
```

✅ Good Documentation (Standard-Compliant)

```rust
/// Duplicates an open file descriptor onto another index, closing the target if active.
///
/// Conforms to POSIX `dup2` semantics. If `oldfd == newfd` and `oldfd` is a valid open
/// descriptor, the function returns `Ok(newfd)` without closing. Otherwise, if `newfd`
/// is already open, it is closed before the clone of `oldfd` is installed.
///
/// If `oldfd` references an anonymous pipe, the associated pipe buffer's reader or writer
/// reference count is incremented accordingly.
///
/// # Arguments
///
/// * `oldfd` - The source file descriptor index (must be in `0..MAX_FDS` and open).
/// * `newfd` - The target file descriptor index to overwrite (must be in `0..MAX_FDS`).
///
/// # Returns
///
/// * `Ok(newfd)` - The new file descriptor index on successful duplication.
/// * `Err("Bad file descriptor (EBADF)")` - If either index is out of bounds or `oldfd` is closed.
///
/// # Errors
///
/// Returns an error if `oldfd ≥ 16`, `newfd ≥ 16`, or `fds[oldfd]` is `None`.
pub fn dup2(&mut self, oldfd: usize, newfd: usize) -> Result<usize, &'static str> {
    // ...
}
```

Example 2: Unsafe Hardware Architecture Function

❌ Bad Documentation

```rust
// switches cr3 page table
pub unsafe fn switch_cr3(phys_addr: u64) {
    // ...
}
```

✅ **Good Documentation (Standard-Compliant)**

```rust
/// Loads a new physical base address into the CPU Control Register 3 (`CR3`).
///
/// Flushes the architectural Translation Lookaside Buffer (TLB) for all non-global
/// page mappings and immediately switches the active 4-level paging root to the
/// specified PML4 table.
///
/// # Arguments
///
/// * `phys_addr` - Physical address of the root PML4 table. Must be strictly 4 KiB aligned
///   (bits 0..11 must be zero).
///
/// # Safety
///
/// This function is inherently unsafe because writing an invalid address to `CR3`
/// causes an immediate Triple Fault and hardware CPU reboot. The caller must guarantee:
/// 1. `phys_addr` points to a valid, identity-mapped or kernel-accessible PML4 frame.
/// 2. The new page table identity-maps the executing code and stack frames.
/// 3. The address satisfies 4096-byte alignment (`phys_addr & 0xFFF == 0`).
///
/// # Hardware Registers
///
/// - Modifies: `CR3` (PDBR - Page Directory Base Register).
/// - Side Effects: Complete invalidation of CPU TLB entries (except those marked Global in PTE).
pub unsafe fn switch_cr3(phys_addr: u64) {
    // ...
}
```

---

8. Verification & Tooling Workflow

Documentation completeness is verified automatically as part of the continuous build:

1. **Local Documentation Generation**: `bash cargo doc --no-deps --target x86_64-unknown-uefi`
2. **Enforcing Warnings as Errors**: Add to crate root (`src/main.rs`): `rust #![warn(missing_docs)] #![warn(rustdoc::broken_intra_doc_links)]`
3. **PDF Generation Synchronization**: Whenever a new module is documented, ensure it is added to `DOCS_ORDER` in `scripts/generate_docs_pdf.py` so it renders into the publication PDF: `bash python scripts/generate_docs_pdf.py`

BUILD & TOOLCHAIN • BOOT SEQUENCE ORDER #03

Build Script Documentation

- **Source File**: `build.rs`
- **Subsystem**: Toolchain & Assembly Build Automation

---

1. Purpose & Context

In earlier phases or legacy configurations, Hydra supported a hybrid BIOS/Multiboot bootloader mode that required compiling 16-bit and 32-bit assembly files (`header.asm` , `main.asm` , `main64.asm`) with the `nasm` assembler before linking into ELF objects.

In current native UEFI builds, Cargo compiles Rust directly targeting `x86_64-unknown-uefi` without requiring external assembly compilation, with `build = false` in `Cargo.toml` .

---

2. Key Functions

`assemble_file(file: &str, output_dir: &str)`

1. Checks the `DEBUG` environment variable to determine whether to emit DWARF debug symbols (`-g`).
2. Invokes `nasm` to compile an `.asm` source file into an ELF64 object (`.o`).
3. Instructs Cargo to re-run the build script if the source assembly file changes (`cargo:rerun-if-changed`).
4. Tells the Rust linker to link the resulting object file (`cargo:rustc-link-object`).

`assemble_all(output_dir: &str)`

Loops through the standard assembly boot files: - `boot/header.asm` : Multiboot2 header definition. - `boot/main.asm` : 32-bit protected mode entry point, page table setup, and long mode switch. - `boot/main64.asm` : 64-bit trampoline to Rust entry point.

STAGE 0: FIRMWARE HANDOFF • BOOT SEQUENCE ORDER #04

Legacy Assembly Bootstrapper Documentation

- **Source Directory**: `src/arch/x86_64/boot/`
- **Files**:
- `header.asm`
- `main.asm`
- `main64.asm`
- **Subsystem**: Architecture / Legacy BIOS Boot Trampoline

---

1. Overview

Before migrating to native **UEFI**, x86 PCs booted via legacy **BIOS**. Under BIOS: 1. The CPU boots in **16-bit Real Mode**. 2. A bootloader (like GRUB) loads the kernel into **32-bit Protected Mode**. 3. The kernel itself is responsible for setting up initial page tables, enabling PAE, switching to **64-bit Long Mode**, and jumping to 64-bit code.

These assembly files implement that transition for reference or hybrid BIOS/GRUB boot media.

---

2. File Breakdown

1. `header.asm` (Multiboot2 Header)

- Defines the magic numbers and architecture tags required by Multiboot2-compliant bootloaders (such as GRUB).
- Allows GRUB to recognize the kernel file as bootable.

2. `main.asm` (32-Bit to 64-Bit Transition)

- **Stack Initialization**: Allocates an initial boot stack.
- **CPU Checks**:
- `check_multiboot`: Verifies that the bootloader passed the expected Multiboot magic value in `EAX`.
- `check_cpuid`: Verifies that the CPU supports the `cpuid` instruction.
- `check_long_mode`: Queries CPUID to verify 64-bit Long Mode hardware support.
- **Identity Paging**: Configures an initial 2 MiB identity-mapped page table so memory addresses map 1:1 during the switch.
- **Enabling Long Mode**: Sets the Physical Address Extension bit (`PAE`) in `CR4`, enables Long Mode in the Extended Feature Enable Register (`EFER` MSR), and enables Paging in `CR0`.
- **GDT Setup**: Loads a temporary 64-bit code segment and performs a far jump to `long_mode_start`.

3. `main64.asm` (64-Bit Trampoline)

- Re-zeros data segment registers (`DS` , `ES` , `FS` , `GS` , `SS`).

- Invokes the Rust 64-bit entry point (`call kernel_main`).
- Halts the CPU if the function ever returns.

STAGE 1: KERNEL ENTRY POINT • BOOT SEQUENCE ORDER #05

Kernel Entry Point & Bootstrapping Documentation

- **Source File**: `src/main.rs`
- **Subsystem**: Microkernel Bootstrap & Initialization Orchestrator

---

1. Conceptual Overview

`src/main.rs` is the primary entry point executed when the UEFI firmware finishes hardware Power-On Self-Test (POST) and transfers control to `BOOTX64.EFI`.

Because Hydra targets native 64-bit UEFI directly: - The CPU is already in **64-bit Long Mode**. - Paging is enabled by the UEFI firmware. - The kernel receives a valid `uefi::prelude::SystemTable` reference, enabling it to access firmware services before taking full ownership of the machine.

---

2. Boot Execution Pipeline

```
graph TD
    A[UEFI Firmware POST] ⟶ B[main entry point]
    B ⟶ C[Initialize Console & Clear Screen]
    C ⟶ D[arch::gdt::init: Load GDT, TSS, Ring 0/3 Segments]
    D ⟶ E[arch::cpu::read_state: Inspect CR0, CR3 PML4 Root, CR4]
    E ⟶ F[memory::frame_allocator: Parse UEFI Memory Map & Test Frame]
    F ⟶ G[memory::paging: Test PML4 Virtual-to-Physical Translation]
    G ⟶ H[capability: Setup Task CSpace & Capability Table]
    H ⟶ I[ipc: Test Endpoint Message Rendezvous]
    I ⟶ J[Halt Loop with Interrupts Disabled]
```

---

3. Step-by-Step Code Walkthrough

The Panic Handler

```
#[panic_handler]
fn panic(_info: &PanicInfo) -> ! {
    console::set_color(Color::LightRed, Color::Black);
    println!("\n[!] KERNEL PANIC: {:?}", _info);
    x86_64::instructions::interrupts::disable();
    loop {
        x86_64::instructions::hlt();
    }
}
```

If an assertion fails or an unexpected unrecoverable condition occurs, the panic handler prints diagnostic error messages in red to the UEFI console, disables interrupts to prevent re-entrancy, and halts the CPU in a low-power infinite loop (`hlt`).

Step 1: Segmentation & Hardware Privilege Setup

```
console::set_color(Color::Yellow, Color::Black);
println!("[*] Step 1: Initializing GDT, TSS & Ring 3 Segments...");
arch::gdt::init();
console::set_color(Color::LightGreen, Color::Black);
println!("[+] GDT loaded successfully.");
```

Disables interrupts and installs Hydra's custom Global Descriptor Table and Task State Segment, configuring the CPU for both supervisor (Ring 0) and isolated userland (Ring 3) execution.

Step 2: CPU Diagnostics

```
let cpu = arch::cpu::read_state();
println!("    - CR0: {:?}", cpu.cr0);
println!("    - CR3 (Active PML4 Root): {:#x}", cpu.cr3_frame.as_u64());
println!("    - CR4: {:?}", cpu.cr4);
```

Reads control registers to log the active hardware configuration, including the physical address of the initial root page table (`CR3`).

Step 3: Physical Memory Discovery & Frame Allocation

```
console::set_color(Color::Yellow, Color::Black);
println!("\n[*] Step 2: Discovering Memory & Initializing Frame Allocator...");
memory::frame_allocator::init_memory_from_uefi();

let (usable_mb, allocated_kb) = {
    let alloc = memory::frame_allocator::FRAME_ALLOCATOR.lock();
    (
        alloc.total_usable_memory() / (1024 * 1024),
        alloc.allocated_memory() / 1024,
    )
};
```

Queries UEFI firmware for the memory map, identifies all conventional RAM regions (skipping the first 1 MiB to protect BIOS/IVT data and null pointers), and initializes the kernel's physical frame allocator. It then tests allocating a sample 4 KiB physical frame.

Step 4: Virtual Memory Verification

```
console::set_color(Color::Yellow, Color::Black);
println!("\n[*] Step 3: Verifying 4-Level Paging (PML4) Address Translation...");
let address_space = memory::paging::AddressSpace::active();
let sample_virt = VirtAddr::new(main as *const () as usize as u64);
let sample_phys = unsafe { address_space.translate_addr(sample_virt) };
```

Walks the active 4-level PML4 page table hierarchy to verify that virtual addresses successfully resolve to physical memory frames, accommodating 4 KiB, 2 MiB, and 1 GiB page mappings.

Step 5: Capability Table Initialization

```
console::set_color(Color::Yellow, Color::Black);
println!("\n[*] Step 4: Initializing Task Capability Space ... ");
let mut cap_table = CapabilityTable::new();
let root_cap = Capability::new(
    CapType::AddressSpace {
        root_cr3: cpu.cr3_frame.as_u64(),
    },
    Rights::ALL,
);
let slot0 = cap_table.insert(root_cap).unwrap();
let ep_cap = Capability::new(
    CapType::Endpoint { endpoint_id: 1 },
    Rights(Rights::READ.0 | Rights::WRITE.0 | Rights::CALL.0),
);
let slot1 = cap_table.insert(ep_cap).unwrap();
```

Establishes the process capability table (CSpace), assigning granular access tokens to kernel resources (Root CR3 address space and IPC Endpoint #1).

Step 6: Microkernel IPC Rendezvous

```
console::set_color(Color::Yellow, Color::Black);
println!("\n[*] Step 5: Testing Microkernel IPC Rendezvous ... ");
let mut endpoint = Endpoint::new(1);
let msg = Message::with_args(0x1001, [42, 1337, 0, 0]).with_cap(ep_cap);
endpoint.send(msg).unwrap();
if let Some(received) = endpoint.receive() { ... }
```

Demonstrates inter-process communication: sending a message with inline arguments and an attached delegated capability, followed by receiving and consuming the message from the endpoint queue.

Step 7: Subsystems Operational & Terminal Halt Loop

```
console::set_color(Color::LightCyan, Color::Black);
println!("\n═══════════════════════════════════════════════════════════");
println!(" [+] HYDRA MICROKERNEL PHASE 1 SUBSYSTEMS OPERATIONAL!      ");
println!("═══════════════════════════════════════════════════════════");

x86_64::instructions::interrupts::disable();
loop {
    x86_64::instructions::hlt();
}
```

Upon successful boot and subsystem diagnostics, the kernel enters a halted state awaiting hardware interrupts or process scheduling.

STAGE 2: EARLY DIAGNOSTICS • BOOT SEQUENCE ORDER #06

Console & Diagnostic Output Documentation

- **Source File**: `src/console.rs`
- **Subsystem**: Core I/O & Formatting

---

1. Conceptual Overview

In a bare-metal `#![no_std]` environment, Rust's standard library (`std::io`, `std::println!`) does not exist because there is no underlying operating system yet.

The `console` module bridges Rust's `core::fmt` formatting system directly to UEFI's standard text output protocol (`SimpleTextOutput`), providing thread-safe `print!` and `println!` macros that can be used everywhere across the kernel.

---

2. Architecture & Concurrency

```
Any Kernel Thread / Subsystem
             |
             v
println!("Text: {}", value)
             |
             v
     CONSOLE (Mutex)  ←── Thread-safe synchronization via spin::Mutex
             |
             v
   uefi::system::with_stdout
             |
             v
   Physical Video Display / UEFI Console Screen
```

- `spin::Mutex`: Used instead of standard OS mutexes because interrupts or threads can acquire the lock without requiring OS scheduling services.

---

3. API Reference & Methods

`print!` and `println!` Macros

Drop-in replacements for standard Rust printing macros.

```
print!("Initializing subsystem... ");
println!("Done: {} frames free.", free_count);
```

`console::set_color(foreground: Color, background: Color)`

Changes the active text color and background color on the UEFI screen.

```
use uefi::proto::console::text::Color;

console::set_color(Color::LightGreen, Color::Black);
println!("[+] Operation Succeeded");
```

`console::clear_screen()`

Clears the active screen buffer and resets the cursor to the top-left coordinate `(0, 0)`.

`console::CONSOLE` and `_print`

```
pub static CONSOLE: Mutex<Console> = Mutex::new(Console);

#[doc(hidden)]
pub fn _print(args: fmt::Arguments) {
    let mut console = CONSOLE.lock();
    let _ = console.write_fmt(args);
}
```

`CONSOLE` is a globally accessible, mutex-locked unit struct implementing `core::fmt::Write`. The `print!` and `println!` macros format arguments and invoke `_print`, acquiring the lock to safely serialize output to the UEFI `SimpleTextOutput` protocol.

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #07

Architecture Module Overview

- **Source File**: `src/arch/mod.rs`
- **Subsystem**: Architecture-Specific Primitives (x86_64)

---

1. Role in Hydra OS

The `arch` module isolates all processor-specific and machine-dependent code away from the rest of the generic microkernel.

When Hydra is ported to other CPU architectures (such as ARM64 or RISC-V), only the implementations inside `arch` need to be swapped out.

---

2. Submodules

- `gdt.rs`: Manages the Global Descriptor Table, Ring 0 / Ring 3 segmentation, and Task State Segment (TSS).
- `cpu.rs`: Provides interfaces for reading x86 control registers (`CR0` , `CR2` , `CR3` , `CR4`).

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #08

Global Descriptor Table (GDT) & TSS Documentation

- **Source File**: `src/arch/gdt.rs`
- **Subsystem**: Architecture / CPU Privilege & Segmentation

---

1. Conceptual Overview

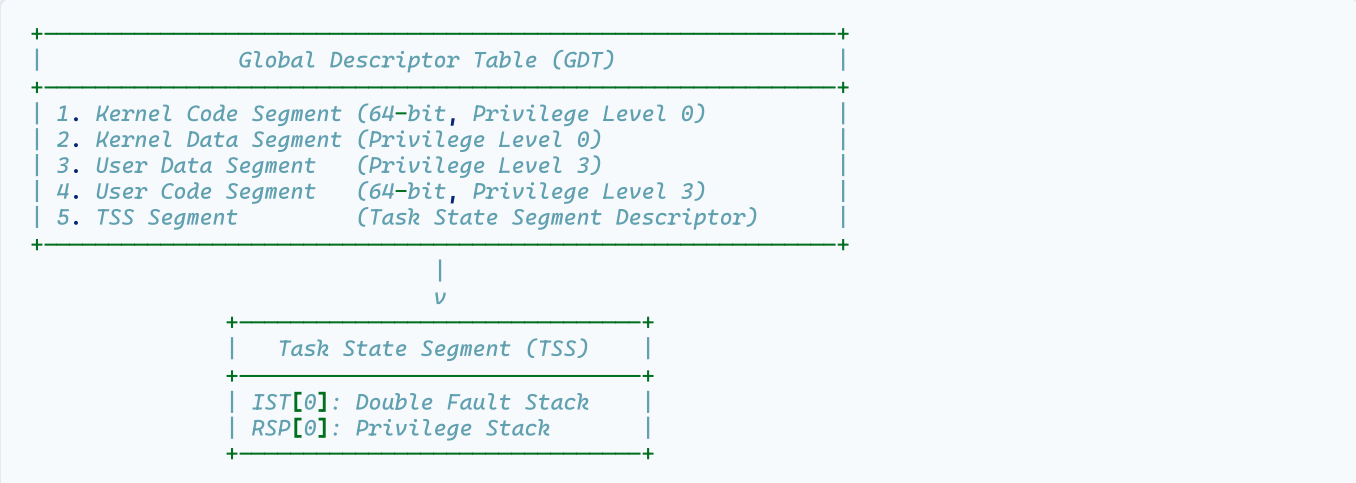
On x86_64 processors, the **Global Descriptor Table (GDT)** is a legacy hardware structure that remains vital for two critical operating system responsibilities: 1. **Privilege Levels (Rings)**: Defining segments for **Ring 0 (Kernel Mode)** and **Ring 3 (Userland Mode)**. 2. **Task State Segment (TSS)**: Defining emergency hardware stack pointers for CPU privilege transitions and catastrophic fault recovery (Double Faults).

Privilege Rings

- **Ring 0**: Full CPU privilege. The kernel executes here and can run privileged instructions (`cli` , `sti` , `invlpg` , write to control registers `CR0..CR4`).
- **Ring 3**: Sandboxed user mode. Applications run here and are restricted by the MMU and capability system.

---

2. Architecture & Components



---

3. Data Structures & Stacks

Stack Allocation

```rust
const STACK_SIZE: usize = 4096 * 5; // 20 KiB
static mut DOUBLE_FAULT_STACK: [u8; STACK_SIZE] = [0; STACK_SIZE];
static mut PRIVILEGE_STACK: [u8; STACK_SIZE] = [0; STACK_SIZE];
```

- `DOUBLE_FAULT_STACK`: When a double fault occurs (e.g. a page fault inside a kernel stack overflow), the CPU cannot use the current corrupted stack. Using the **Interrupt Stack Table (IST index 0)**, the CPU automatically switches to this clean stack.
- `PRIVILEGE_STACK`: When a Ring 3 user program triggers a system call or interrupt to enter Ring 0, the CPU loads this stack into `RSP` (`tss.privilege_stack_table[0]`).

`Selectors`

Stores the segment selectors returned when the segments are appended to the GDT:

```
pub struct Selectors {
    pub kernel_code_selector: SegmentSelector,
    pub kernel_data_selector: SegmentSelector,
    pub user_data_selector: SegmentSelector,
    pub user_code_selector: SegmentSelector,
    pub tss_selector: SegmentSelector,
}
```

---

4. API Reference & Methods

`arch::gdt::init()`

Initializes and prepares Hydra's segmentation and hardware privilege structures in memory: 1. **Prepares Task State Segment (TSS)**: Configures `interrupt_stack_table[0]` for the double-fault emergency stack and `privilege_stack_table[0]` for Ring 3 → Ring 0 privilege transitions. 2. **Builds the GDT**: Allocates and configures Kernel Code, Kernel Data, User Data, User Code, and TSS descriptors. 3. **UEFI Firmware Coexistence**: While native UEFI Boot Services are active, the firmware's IDT relies on firmware code selectors (e.g., `0x38` on VirtualBox EDK2). Hydra ensures GDT and TSS structures are initialized and accessible via `get_selectors()` without disrupting the firmware's active interrupt vectors.

```
arch::gdt::init();
```

`arch::gdt::get_selectors() → &'static Selectors`

Provides access to the active segment selectors (`kernel_code_selector` , `kernel_data_selector` , `user_data_selector` , `user_code_selector` , `tss_selector`) for subsequent userland process spawning, context switches, and interrupt returns.

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #09

Interrupt Descriptor Table (IDT) & Hardware Exceptions

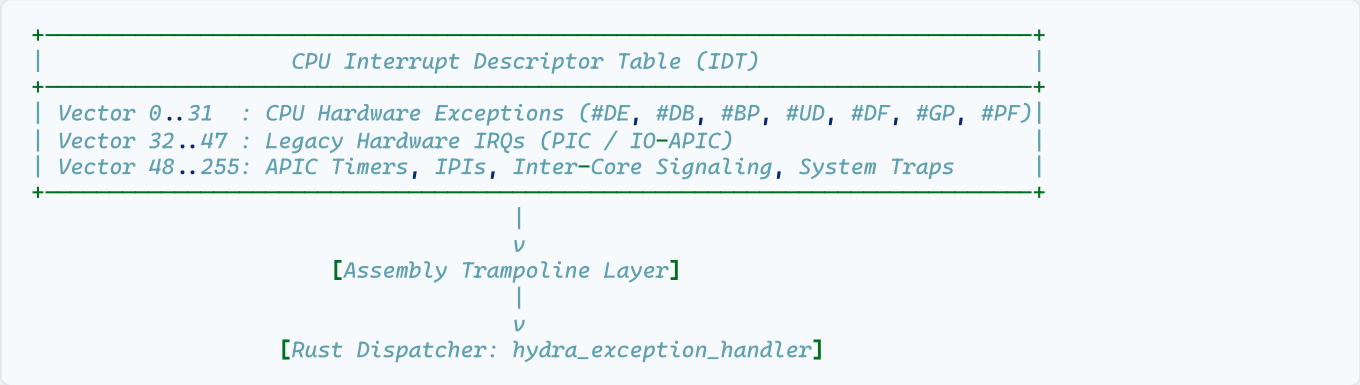
- **Source File**: `src/arch/idt.rs`
- **Subsystem**: Architecture / CPU Exception Handling & Interrupt Routing

---

1. Conceptual Overview

The **Interrupt Descriptor Table (IDT)** is the x86_64 CPU hardware mechanism that dictates how the processor responds to asynchronous interrupts and synchronous software/hardware exceptions.

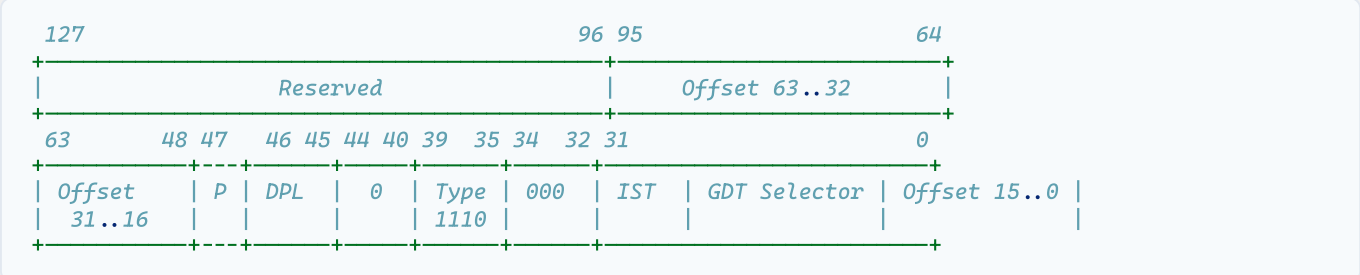
Each entry in the table is a 16-byte gate descriptor defining: 1. **Target Handler Address**: The 64-bit instruction pointer (`RIP`) to jump to when the vector fires. 2. **Segment Selector**: The GDT code segment selector (Hydra Ring 0 Kernel Code Segment). 3. **Gate Type & DPL**: Descriptor Privilege Level (DPL 0 for kernel-only hardware exceptions; DPL 3 for userland-callable traps like `int3`). 4. **Interrupt Stack Table (IST)**: Optional alternate stack pointer index (1..=7) in the TSS, ensuring critical faults execute on guaranteed uncorrupted stack frames.



---

2. 64-Bit Interrupt Gate Structure

In Long Mode (64-bit), each IDT descriptor is 16 bytes:



Hydra models this with `#[repr(C, packed)] pub struct IdtEntry` :

```
#[repr(C, packed)]
pub struct IdtEntry {
    offset_low: u16,
    selector: u16,
    ist: u8,
    type_attr: u8,
    offset_mid: u16,
    offset_high: u32,
    reserved: u32,
}
```

---

3. Trampoline Layer & Calling Convention

Hardware Exception Frame Variations

When a CPU exception occurs: - Some exceptions push an **Error Code** automatically (`#DF` , `#TS` , `#NP` , `#SS` , `#GP` , `#PF` , `#AC`). - Other exceptions push **no Error Code** (`#DE` , `#DB` , `#BP` , `#OF` , `#UD` , `#NM`).

To provide a uniform C-compatible context to Rust, assembly macros dynamically push a dummy zero for exceptions that lack hardware error codes:

```
.macro TRAP_NO_ERRCODE name, vec
.global \name
\name:
    push 0                  // Dummy error code
    push \vec               // Vector number
    jmp common_trap_handler
.endm

.macro TRAP_ERRCODE name, vec
.global \name
\name:
    push \vec               // Vector number (error code already pushed by CPU)
    jmp common_trap_handler
.endm
```

Stack Frame (`ExceptionContext`)

The assembly trampoline pushes all 15 general-purpose registers, creating a complete `ExceptionContext` :

```
#[repr(C)]
pub struct ExceptionContext {
    pub r15: u64, pub r14: u64, pub r13: u64, pub r12: u64,
    pub rbp: u64, pub rbx: u64, pub r11: u64, pub r10: u64,
    pub r9:  u64, pub r8:  u64, pub rdi: u64, pub rsi: u64,
    pub rdx: u64, pub rcx: u64, pub rax: u64,
    pub vector: u64,
    pub error_code: u64,
    pub rip: u64, pub cs: u64, pub rflags: u64, pub rsp: u64, pub ss: u64,
}
```

UEFI & SysV Calling Convention Compatibility

Because Hydra targets UEFI (`x86_64-unknown-uefi`) while maintaining Unix microkernel foundations, the dispatcher assembly sets up argument passing for both conventions simultaneously:

```
    mov rcx, rsp            // 1st argument in Microsoft x64 / UEFI ABI
    mov rdi, rsp            // 1st argument in System V AMD64 ABI
    sub rsp, 32             // Allocate 32-byte shadow store (Win64) & align stack
    call hydra_exception_handler
    add rsp, 32             // Deallocate shadow store
    pop r15..rax            // Restore registers
    add rsp, 16             // Clean up vector and error code
    iretq                   // Return from interrupt
```

---

4. Exception Handlers

Vector 3: Breakpoint (`#BP`)

- **DPL**: 3 (Permits Ring 3 userland processes to trigger software breakpoints with `int3`).
- **Behavior**: Non-fatal diagnostic trap. Prints RIP, RFLAGS, and RSP, then resumes execution via `iretq`.

Vector 8: Double Fault (`#DF`)

- **IST**: Linked to TSS IST index 1 (`DOUBLE_FAULT_STACK`).
- **Behavior**: Even if the kernel stack overflows or suffers memory corruption, the hardware forcibly switches to a dedicated 20 KiB emergency stack, preventing immediate triple-fault reboot loops.

Vector 13: General Protection Fault (`#GP`)

- **Behavior**: Decodes selector indices and privilege boundary violations.

Vector 14: Page Fault (`#PF`)

- **Behavior**: Reads the faulting linear memory address from hardware control register `CR2` (`Cr2::read_raw()`), and decodes the error code flags:
- **P (Bit 0)**: Non-present page (0) vs Page-level protection violation (1).
- **W/R (Bit 1)**: Read access (0) vs Write access (1).
- **U/S (Bit 2)**: Supervisor mode (0) vs User mode (1).
- **I/D (Bit 4)**: Data access (0) vs Instruction fetch (1).

---

5. Shell Integration & Diagnostics

Hydra includes a built-in shell diagnostic command `trap` (aliased to `int3` and `breakpoint`):

```
hydra:/> trap
[*] Executing INT3 instruction to test IDT breakpoint trap...

[i] CPU BREAKPOINT TRAP (#BP) at RIP: 0x1f07f439
    RFLAGS: 0x246, RSP: 0x1ee15bf0
[+] Resumed successfully from #BP exception handler!
```

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #10

8259 Programmable Interrupt Controller (PIC)

1. Architectural Role

The **8259 Programmable Interrupt Controller (PIC)** is the legacy interrupt routing hardware in standard PC/AT architectures. It aggregates up to 15 hardware IRQs (Interrupt Request lines) across two cascading controllers and forwards them to the CPU via the `INTR` line.

In standard x86 BIOS systems, hardware IRQs initially collide with CPU exception vectors (`0x00–0x1F`). Hydra remaps the dual-PIC controllers to vectors `0x20–0x2F` upon hardware takeover (`exitboot`), ensuring clear separation between CPU faults and hardware peripherals.

---

2. Port & Vector Mapping

Controller	Command Port	Data Port	Vector Range	IRQ Lines
Master PIC	`0x20`	`0x21`	`0x20..0x27` (32..39)	IRQ 0..7 (Timer, PS/2 Keyboard, Cascade, etc.)
Slave PIC	`0xA0`	`0xA1`	`0x28..0x2F` (40..47)	IRQ 8..15 (RTC, ACPI, Primary/Secondary ATA)

---

3. Initialization & Remapping Sequence

Initialization uses a sequence of 4 Initialization Command Words (**ICWs**):

```
ICW1 (0x11): Start initialization in cascade mode, expect ICW4
    │
    ├─→ ICW2: Set base vector offset
    │     Master = 0x20 (32)
    │     Slave  = 0x28 (40)
    │
    ├─→ ICW3: Cascade wiring setup
    │     Master: bit 2 = 1 (Slave attached at IRQ 2)
    │     Slave : cascade identity 0x02
    │
    └─→ ICW4 (0x01): 8086/88 operation mode
```

---

4. Interrupt Masking & End of Interrupt (EOI)

- **Default Masks**:
- `Master Mask` : `0xF8` (Unmasks IRQ 0 Timer, IRQ 1 Keyboard, IRQ 2 Cascade).
- `Slave Mask` : `0xFF` (All slave IRQs masked).
- **End of Interrupt (EOI)**:

- When an interrupt is serviced, an EOI byte (`0x20`) must be written to the PIC command port.
- If the IRQ originated from the Slave PIC (`irq ≥ 8`), an EOI must be sent to **both** Slave (`0xA0`) and Master (`0x20`).

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #11

8254 Programmable Interval Timer (PIT)

1. Architectural Role

The **Intel 8254 Programmable Interval Timer (PIT)** provides Hydra with a hardware clock source for measuring wall-clock time, coordinating preemptive multitasking ticks, and executing low-power sleeps.

The oscillator operates at an input frequency of **1,193,182 Hz** (~1.193182 MHz). Channel 0 is wired directly to IRQ 0 on the Master PIC.

---

2. Configuration & Ports

- **Port** `0x40`: Channel 0 Data Port (read/write counter divisor)
- **Port** `0x43`: Mode/Command Register

Hydra configures Channel 0 in **Mode 3 (Square Wave Generator)**: - Command Byte: `0x36` (`00` Channel 0 | `11` Access lo/hi byte | `011` Mode 3 | `0` 16-bit binary) - Target Frequency: **100 Hz** (10 ms per tick) - Divisor: `1193182 / 100 = 11932` (`0x2E9C`)

---

3. Tick Counter & Low-Power Halts

On every IRQ 0 (Vector 32) interrupt: 1. `pit::on_timer_tick()` increments a 64-bit atomic counter `TICKS`. 2. The PIC is notified via `pic::send_eoi(0)`. 3. Routines like `sleep_ms()` check `ticks() < target` and execute `hlt` instructions, keeping CPU usage near 0% while waiting.

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #12

8042 PS/2 Keyboard Driver & Input Subsystem

1. Architectural Role

The **8042 Keyboard Controller** provides Hydra with independent, firmware-free human interface input. Operating on standard I/O ports `0x60` (Data) and `0x64` (Status/Command), it decodes raw hardware scancodes into high-level keyboard events.

Following Phase 1, the driver operates in dual mode: - **Interrupt-Driven Mode (Hardware Mode)**: IRQ 1 (IDT Vector 33) captures scancodes asynchronously into a 128-key circular ring buffer (`KeyRingBuffer`), enabling low-power `hlt` execution while waiting for input. - **Polling Fallback**: Reads status bit 0 from port `0x64` when hardware interrupts are not yet active (e.g. early kernel bootstrap).

---

2. Port Architecture

Port	Direction	Function
`0x60`	Read	Data Port (receives raw scancodes from keyboard)
`0x64`	Read	Status Register (Bit 0 = Output Buffer Full / Byte Ready)

---

3. Scancode Set 1 & Modifier Tracking

The driver tracks modifier state machines for: - **Shift (`0x2A` / `0x36`)**: Make/Break tracking alters character case and symbols. - **Ctrl (`0x1D`)**: Translates ASCII letters into control characters `^A` .. `^Z` (ASCII `1..26`) for interrupt signaling (`Ctrl+C` , `Ctrl+S`). - **Extended Prefix (`0xE0`)**: Signals two-byte extended navigation keys (Arrow keys, Home, End, Delete, PageUp, PageDown).

---

4. Circular Ring Buffer & Low-Power Halts



When the interactive shell waits for user input, `get_key()` invokes `x86_64::instructions::hlt()`. The CPU enters a sleep state until the next timer tick (IRQ 0) or keystroke (IRQ 1) wakes the core, reducing idle power consumption to near zero.

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #13

Hardware Takeover & ExitBootServices

- **Source File**: `src/arch/boot_services.rs`
- **Subsystem**: Architecture / Hardware Ownership & UEFI Termination

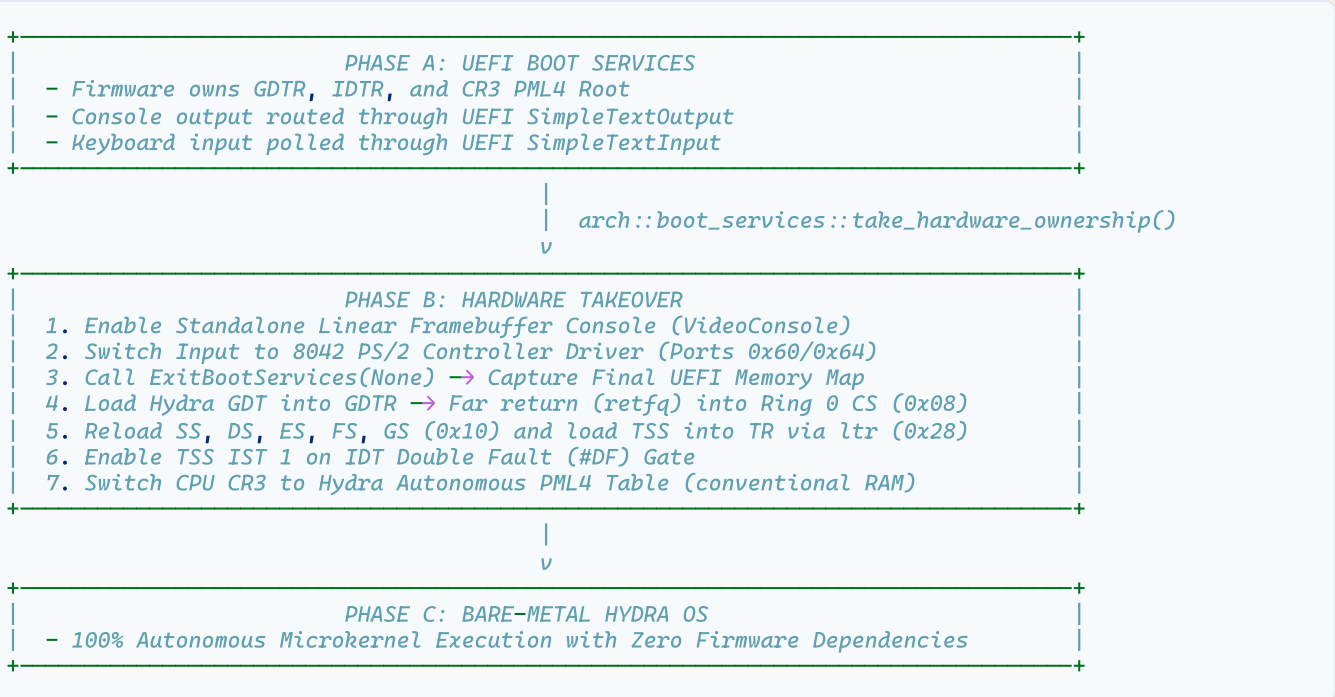
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1. Conceptual Overview

Operating systems that boot via UEFI start their existence in a hybrid environment: the processor is in 64-bit Long Mode, but the firmware retains partial control over memory allocation, display devices, and the system timer via **UEFI Boot Services**.

To become a true, self-governing microkernel, Hydra OS must execute the **Hardware Takeover Sequence** by calling the UEFI specification's terminal function: `ExitBootServices`.

Once `ExitBootServices` succeeds: 1. **Firmware Eviction**: All UEFI boot-time protocols, memory allocators, and event timers cease execution immediately. 2. **Hardware Reconfiguration**: The kernel assumes exclusive control over the Global Descriptor Table (GDTR), Task State Segment (TR), Interrupt Descriptor Table (IDTR), and Page Table Root (CR3). 3. **Autonomous Driver Transition**: Console output routes directly to Hydra's hardware GOP linear framebuffer (`VideoConsole`), and keyboard input routes directly to the physical 8042 PS/2 keyboard controller.



---

2. Hardware Takeover Pipeline

Step 1: Autonomous Video & Keyboard Activation

Before firmware protocols are severed, Hydra switches console routing:

```
enable_standalone_video(true);
set_hardware_mode(true);
```

All subsequent `print!` and `println!` calls draw bitmap glyphs directly into physical video memory (`0x80000000`), and `get_key()` reads scancodes from port `0x60` .

Step 2: Calling `ExitBootServices`

```
let memory_map = unsafe { uefi::boot::exit_boot_services(None) };
let entries_count = memory_map.entries().count();
```

The firmware validates the memory map key, frees internal boot-service structures, and hands unconditional hardware control to Hydra.

Step 3: GDT & Segment Reload (`load_and_switch`)

In 64-bit mode, the Code Segment (`CS`) cannot be reloaded using `mov cs, ax` . Hydra uses a far return (`retfq`) trampoline to reload `CS` with `0x08` :

```
    push {sel}              // Kernel Code Selector (0x08)
    lea {tmp}, [2f + rip]   // Return RIP
    push {tmp}
    retfq                   // Far return: pops RIP then CS
2:
    mov ds, 0x10
    mov ss, 0x10
    ltr {tss_sel}           // Load Task Register with 0x28
```

Step 4: TSS IST Attachment on Double Fault

With the Task Register (`TR`) loaded with Hydra's TSS, the IDT entry for Double Fault (`#DF` , vector 8) is linked to IST index 1:

```
idt.entries[8].set_ist(1);
```

Any catastrophic kernel stack overflow will now switch immediately to Hydra's dedicated 20 KiB emergency stack.

Step 5: Autonomous CR3 Switch

Hydra switches the processor's Page Map Level 4 root from the firmware's PML4 table to Hydra's autonomous PML4 table:

```
KERNEL_SPACE.lock().switch();
```

---

3. The 8042 PS/2 Keyboard Driver

In bare-metal mode, keystrokes are processed by `src/arch/keyboard.rs`: - **Status Port** `0x64`: Polled for bit 0 (`Output Buffer Full`). - **Data Port** `0x60`: Reads Scancode Set 1 make/break codes. - **Modifier Tracking**: Tracks Shift (`0x2A` / `0x36`) and Ctrl (`0x1D`) state to generate uppercase letters, punctuation, and control characters (such as `Ctrl+C`). - **Extended Scancodes (`0xE0`)**: Decodes arrow keys (`Up`, `Down`, `Left`, `Right`), `PageUp`, `PageDown`, `Home`, and `End` for terminal line editing and history navigation.

---

4. Interactive Verification & Shell Command

The microkernel includes an interactive shell command `exitboot` (aliased to `takeover`, `baremetal`, and `hwmode`):

```
hydra:/bin> exitboot status
[-] Hardware Mode: INACTIVE (Running under UEFI Boot Services)
    Type 'exitboot' to terminate boot services and take full hardware control.

hydra:/bin> exitboot
[*] Initiating ExitBootServices and Bare-Metal Hardware Takeover...

================================================================
 [+] HYDRA MICROKERNEL: FULL HARDWARE TAKEOVER COMPLETE!
================================================================
  - UEFI Boot Services : TERMINATED (ExitBootServices invoked)
  - Memory Map Entries : 61 descriptors captured
  - GDT & TSS Status   : Hydra Ring 0 CS=0x8, DS=0x10, TR=0x28
  - Interrupts & IDT   : Hydra 256 gates (Double Fault on TSS IST1)
  - Virtual Memory     : Autonomous PML4 Root at PhysAddr 0x901000
  - Display & Input    : 1024x768 Linear Framebuffer & 8042 PS/2 Keyboard
================================================================
```

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #14

CPU Control & Register Diagnostics Documentation

- **Source File**: `src/arch/cpu.rs`
- **Subsystem**: Architecture / CPU Control Registers

---

1. Conceptual Overview

On x86_64 processors, the processor's operating mode, memory translation mechanisms, and security features are governed by specialized **Control Registers** (`CR0` , `CR2` , `CR3` , `CR4`):

- **CR0**: Controls basic execution features: Protected Mode, Paging enable, Write Protect (prevents kernel from inadvertently overwriting read-only user pages).
- **CR2**: Stores the linear virtual address that triggered the most recent **Page Fault** exception. Vital for demand paging and memory debugging.
- **CR3**: Points to the base physical address of the Level 4 Page Table (PML4). Loading a new value into CR3 immediately switches active virtual address spaces.
- **CR4**: Controls advanced architecture features: Physical Address Extension (PAE), Page Global Enable (PGE), OS support for SSE/AVX floating point, and User-Mode Instruction Prevention (UMIP).

---

2. The `CpuState` Struct

```
#[derive(Debug, Clone, Copy)]
pub struct CpuState {
    pub cr0: Cr0Flags,
    pub cr2: u64,
    pub cr3_frame: PhysAddr,
    pub cr4: Cr4Flags,
}
```

Captures an instantaneous snapshot of the machine's hardware configuration registers.

---

3. API Reference

`arch::cpu::read_state() → CpuState`

Queries the CPU hardware registers and packages them into a `CpuState` diagnostic struct.

```
use crate::arch::cpu;

let cpu = cpu::read_state();
println!("Paging root physical address: {:#x}", cpu.cr3_frame.as_u64());
```

STAGE 3: CPU ARCHITECTURE • BOOT SEQUENCE ORDER #15

Ring 3 User Mode & Hardware System Calls (`syscall` / `sysretq`)

Overview

Phase 3 introduces hardware-enforced CPU privilege rings and the fast system call interface to Hydra OS. In long mode x86_64, the kernel transitions from pure kernel-mode threads (Ring 0) to unprivileged user processes executing at Current Privilege Level 3 (**Ring 3**).

The boundary between userland and the microkernel is guarded by: 1. **Model-Specific Registers (MSRs)**: `IA32_EFER.SCE`, `IA32_STAR`, `IA32_LSTAR`, and `IA32_FMASK` configuring the hardware `syscall` and `sysretq` instructions. 2. **Task State Segment (TSS) Privilege Stack (`RSP0`)**: Ensuring hardware interrupts and exceptions originating from Ring 3 safely pivot to a trusted per-task kernel stack. 3. **4-Level Paging Protection (`USER_ACCESSIBLE`)**: Granular page table permissions separating supervisor-only memory from user-accessible memory. 4. **Hardware Fault Isolation**: Intercepting user-mode General Protection Faults (#GP) and Page Faults (#PF) to terminate offending processes without compromising kernel stability.

The core implementation is split across: - **Syscall MSRs & Assembly Dispatcher**: `src/arch/syscall.rs` - **Userland Syscall Library**: `src/syscall/user.rs` - **GDT & TSS RSP0 Configuration**: `src/arch/gdt.rs` - **Page Table User Accessibility**: `src/memory/paging.rs` - **Process Spawning & Preemption**: `src/task/scheduler.rs` - **Shell Diagnostics**: `src/shell/commands/task_cmds.rs`

Hardware System Call Architecture

```
sequenceDiagram
    autonumber
    actor User as Ring 3 Process
    participant CPU as x86_64 CPU Hardware
    participant Stub as asm_syscall_entry (Ring 0)
    participant Disp as hydra_syscall_dispatcher
    participant Sched as Hydra Scheduler / Kernel

    User->>CPU: syscall (RAX=num, RDI=arg0, RSI=arg1, RDX=arg2)
    Note over CPU: Saves RIP to RCX, RFLAGS to R11<br/>Loads CS=0x08, SS=0x10<br/>RFLAGS &= ~FMASK (IF
    CPU->>Stub: Enters asm_syscall_entry
    Note over Stub: Swaps RSP to CURRENT_KERNEL_RSP<br/>Constructs SyscallFrame on kernel stack
    Stub->>Disp: hydra_syscall_dispatcher(&mut SyscallFrame)
    Disp->>Sched: Dispatches SYS_WRITE, SYS_SLEEP, SYS_EXIT, etc.
    Sched-->>Disp: Result / Status Code
    Disp-->>Stub: Updates frame.rax with return value
    Note over Stub: Restores GPRs, restores User RSP
    Stub->>CPU: sysretq
    Note over CPU: Loads CS=0x23, SS=0x1B<br/>Restores RIP from RCX, RFLAGS from R11<br/>Switches to R
    CPU-->>User: Resumes execution in user mode
```

Model-Specific Registers (MSR) Configuration

During hardware takeover in `src/arch/boot_services.rs`, the kernel programs the four architectural system call MSRs via `crate::arch::syscall::init()`:

MSR Address	Register Name	Configured Value	Description
`0xC0000080`	`IA32_EFER`	Bit 0 (`SCE = 1`)	**System Call Extensions**: Enables the `syscall` and `sysretq` instructions in 64-bit mode.
`0xC0000081`	`IA32_STAR`	`0x0013_0008_0000_0000`	**Segment Selectors**: Bits [47:32] set Kernel CS to `0x08` (Kernel SS auto `0x10`). Bits [63:48] set User base to `0x13` , causing `sysretq` to load User SS = `0x1B` (0x13+8) and User CS = `0x23` (0x13+16).
`0xC0000082`	`IA32_LSTAR`	Virtual address of `asm_syscall_entry`	**Target Instruction Pointer**: Canonical entry point where the CPU jumps upon executing `syscall` .
`0xC0000084`	`IA32_FMASK`	`0x0004_4700`	**Flags Mask**: Clears Interrupt Flag (IF: `0x200`), Trap Flag (TF: `0x100`), Direction Flag (DF: `0x400`), Nested Task (NT: `0x4000`), and Alignment Check (AC: `0x40000`) on syscall entry.

Assembly Syscall Trampoline (`asm_syscall_entry`)

Because the x86_64 `syscall` instruction does **not** switch the stack pointer (`RSP`), the CPU enters kernel mode with `RSP` still pointing into untrusted user space. The assembly trampoline immediately pivots to the task's kernel stack before saving registers:

```
.global asm_syscall_entry
asm_syscall_entry:
    // 1. Preserve user RSP in scratch and switch to current task kernel stack
    mov [rip + USER_RSP_SCRATCH], rsp
    mov rsp, [rip + CURRENT_KERNEL_RSP]

    // 2. Build SyscallFrame on kernel stack
    push qword ptr [rip + USER_RSP_SCRATCH] // User RSP
    push r11                                // User RFLAGS
    push rcx                                // User RIP
    push rbx
    push rbp
    push r12
    push r13
    push r14
    push r15
    push rax                                // Syscall number
    push rdi                                // Arg 0
    push rsi                                // Arg 1
    push rdx                                // Arg 2
    push r10                                // Arg 3
    push r8                                 // Arg 4
    push r9                                 // Arg 5

    // 3. Call Rust dispatcher (passes &mut SyscallFrame in RCX and RDI)
    mov rcx, rsp
    mov rdi, rsp
    sub rsp, 48
    call hydra_syscall_dispatcher
    add rsp, 48

    // 4. Restore registers and user stack
    pop r9
    pop r8
    pop r10
    pop rdx
    pop rsi
    pop rdi
    pop rax                                 // Syscall return value in RAX
    pop r15
    pop r14
    pop r13
    pop r12
    pop rbp
    pop rbx
    pop rcx                                 // Restore User RIP
    pop r11                                 // Restore User RFLAGS
    pop rsp                                 // Restore User RSP directly

    // 5. Return to unprivileged Ring 3
    sysretq
```

Task State Segment (TSS) & Interrupt Stack Switching

When a hardware interrupt (such as IRQ 0 PIT timer or IRQ 1 keyboard) or a CPU exception (#GP, #PF) fires while running in Ring 3: 1. The CPU hardware reads `RSP0` from the active TSS (`privilege_stack_table[0]`). 2. The hardware switches `RSP` to `RSP0` . 3. The hardware pushes `SS` (0x1B), `RSP` (user stack), `RFLAGS` , `CS` (0x23), `RIP` , and error code onto the task's kernel stack. 4. The interrupt routine runs safely in Ring 0. 5. Upon returning, `iretq` automatically pops the user context and drops back to Ring 3.

Whenever the scheduler performs a context switch in `src/task/scheduler.rs` , it synchronizes both hardware stack pointers:

```
// Update hardware TSS RSP0 and SYSCALL kernel RSP for the incoming task
let kernel_top = next_task.kernel_stack_top();
crate::arch::gdt::set_privilege_stack(x86_64::VirtAddr::new(kernel_top));
crate::arch::syscall::set_kernel_rsp(kernel_top);
```

Registered System Calls

Hydra OS provides a microkernel system call table:

Syscall #	Name	Signature	Description
1	SYS_YIELD	sys_yield()	Cooperatively relinquishes the current timeslice to the next ready task.
2	SYS_SLEEP	sys_sleep(ms: u64)	Suspends the process for a duration in milliseconds.
3	SYS_WRITE	sys_write(buf: *const u8, len: u64)	Writes a UTF-8 character buffer directly to the linear framebuffer console.
4	SYS_EXIT	sys_exit() -> !	Permanently terminates the calling process and yields CPU ownership.
5	SYS_GETPID	sys_getpid() -> u64	Queries the process identifier (PID) of the calling task.
6	SYS_UPTIME	sys_uptime() -> u64	Queries elapsed PIT timer ticks (100 Hz = 10 ms per tick).
7	SYS_IPC_SEND	sys_ipc_send(ep: u32, msg: *const Message)	Sends an IPC message to a microkernel endpoint.
8	SYS_IPC_RECV	sys_ipc_recv(ep: u32, msg: *mut Message)	Receives an IPC message from a microkernel endpoint.

Memory Protection & Fault Isolation

CR0.WP Page Table Manipulation (`make_user_accessible`)

Under UEFI firmware, page tables are marked read-only with CR0.WP (Write Protect) enabled. When allocating user stacks on the heap, the kernel clears CR0.WP temporarily to mark intermediate page table entries (PML4, PDPT, PD, PT) with `PageTableFlags::USER_ACCESSIBLE | PageTableFlags::WRITABLE`:

```
// Temporarily clear CR0.WP (bit 16)
core::arch::asm!(
    "mov {tmp}, cr0",
    "btr {tmp}, 16",
    "mov cr0, {tmp}",
    tmp = out(reg) _,
    options(nostack, preserves_flags),
);
// Mark hierarchy levels USER_ACCESSIBLE
...
// Re-enable CR0.WP
core::arch::asm!(
    "mov {tmp}, cr0",
    "bts {tmp}, 16",
    "mov cr0, {tmp}",
    tmp = out(reg) _,
    options(nostack, preserves_flags),
);
```

Process Isolation on Hardware Faults

If a user process performs an illegal operation (e.g. executing `cli`, `hlt`, `in`, `out`, or accessing invalid memory): - **General Protection Fault (#GP, Vector 13)**: Intercepted in `src/arch/idt.rs`. If `(ctx.cs & 3) == 3`, the kernel reports the fault address, kills only the offending task (`sched.terminate_current()`), and immediately switches to another ready task. - **Page Fault (#PF, Vector 14)**: If `(ctx.error_code & 4) ≠ 0` (user-mode fault), the kernel reports a Segmentation Fault, kills the task, and maintains system integrity. - **The interactive shell and other tasks remain completely unaffected.**

Interactive Shell Verification

Three shell commands expose and verify Phase 3 functionality:

```
hydra:/bin> syscalls

================================================================
[+] HYDRA MICROKERNEL: HARDWARE SYSTEM CALL SUBSYSTEM
================================================================
Hardware MSR Registers:
  - IA32_EFER.SCE (0xC0000080) : 0x0000000000000d01 (SCE=1)
  - IA32_STAR     (0xC0000081) : 0x0013000800000000
    * Kernel CS:SS Base : 0x0008 / 0x0010
    * User CS:SS Base   : 0x0013 (CS=0x0023, SS=0x001b)
  - IA32_LSTAR    (0xC0000082) : 0x0000000001dd79364 (asm_syscall_entry)
  - IA32_FMASK    (0xC0000084) : 0x0000000000044700 (Flags masked on entry)

Registered System Calls & Invocation Counts:
  - [1] SYS_YIELD    : 0 calls
  - [2] SYS_SLEEP    : 4 calls
  - [3] SYS_WRITE    : 10 calls
  - [4] SYS_EXIT     : 1 calls
  - [5] SYS_GETPID   : 1 calls
  - [6] SYS_UPTIME   : 1 calls
  - [7] SYS_IPC_SEND : 0 calls
  - [8] SYS_IPC_RECV : 0 calls
  ------------------------------------------
  * Total Hardware System Calls Handled: 17
================================================================

hydra:/bin> ps
PID   NAME        MODE    STATE     PRIORITY    TICKS        STACK (RSP)
-----------------------------------------------------------------------------
0     shell       Kernel  Running   Normal       6298 0x0000000001fe5e80
1     idle        Kernel  Ready     Idle            0 0x00000000017a3190
2     user_demo   User    Dead      Normal          8 0x00000000017a8870
3     rogue_task  User    Dead      Normal          2 0x00000000017b6188
-----------------------------------------------------------------------------
Total tasks: 4 | Timeslice: 50 ms | Worker counter: 0
```

STAGE 4: MEMORY SUBSYSTEM • BOOT SEQUENCE ORDER #16

Memory Subsystem Overview

- **Source File**: `src/memory/mod.rs`
- **Subsystem**: Memory Management

---

1. Role in Hydra OS

The `memory` module coordinates all physical and virtual memory resources required by the kernel and user processes.

It abstracts low-level hardware memory mechanisms (UEFI memory descriptors, x86_64 paging tables, MMU control registers) into safe, modular Rust interfaces.

---

2. Submodules

- `frame_allocator.rs`: Manages physical memory frames (4 KiB) discovered from the UEFI firmware memory map.
- `heap.rs`: Dynamic kernel heap allocator and global allocator (`alloc` crate) providing `Box`, `Vec`, `String`.
- `paging.rs`: Manages virtual memory address spaces and implements 4-level PML4 address translation.

STAGE 4: MEMORY SUBSYSTEM • BOOT SEQUENCE ORDER #17

Physical Memory & Frame Allocator Documentation

- **Source File**: `src/memory/frame_allocator.rs`
- **Subsystem**: Memory Management (Physical Layer)

---

1. Conceptual Overview

When a computer boots up, its physical Random Access Memory (RAM) is not one continuous empty space. Certain chunks are occupied by motherboard ROM, video graphics RAM, ACPI tables, and UEFI firmware code.

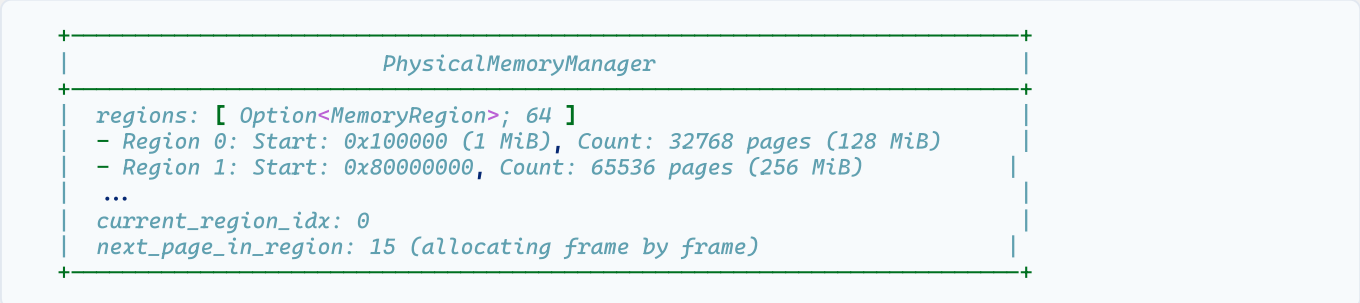
The **Physical Memory Frame Allocator** is responsible for: 1. Asking UEFI firmware which memory regions are safe and usable (`CONVENTIONAL` memory). 2. Dividing that usable RAM into standard **4096-byte (4 KiB) chunks**, called **Physical Frames**. 3. Handing out frames one-by-one to the kernel when it needs memory for page tables, stacks, or user processes.

The Real-World Analogy

Think of RAM as a vast parking lot: - Some spots are reserved for emergency vehicles or building infrastructure (firmware/hardware). - The **Physical Memory Manager** is the parking attendant with a map of all standard open spots. - Whenever a driver (a process or page table) arrives, the attendant assigns them the next available numbered spot (`0x1000` , `0x2000` , etc.).

---

2. Architecture & Data Structures



Key Constants

- `PAGE_SIZE: u64 = 4096` : Standard x86_64 base page size (4 KiB).
- `MAX_REGIONS: usize = 64` : Maximum number of non-contiguous usable RAM blocks tracked.

---

3. Struct Details

`MemoryRegion`

Represents a contiguous block of usable physical memory reported by firmware:

```
#[derive(Debug, Clone, Copy)]
pub struct MemoryRegion {
    pub start: PhysAddr,
    pub page_count: u64,
}
```

`PhysicalMemoryManager`

The state tracker for physical memory:

```
pub struct PhysicalMemoryManager {
    regions: [Option<MemoryRegion>; MAX_REGIONS],
    region_count: usize,
    current_region_idx: usize,
    next_page_in_region: u64,
    total_usable_bytes: u64,
    allocated_frames_count: u64,
}
```

---

4. API Reference & Methods

`PhysicalMemoryManager::add_region(&mut self, start: PhysAddr, page_count: u64)`

Registers a contiguous block of usable physical memory into the tracker.

`PhysicalMemoryManager::allocate_frame(&mut self) -> Option<PhysFrame<Size4KiB>>`

Finds the next free 4 KiB frame from the current memory region and advances the bump pointer. If a region is exhausted, it moves to the next region. - Returns `Some(PhysFrame)` containing the physical start address. - Returns `None` if physical memory is exhausted (Out of Memory).

`PhysicalMemoryManager::total_usable_memory(&self) -> u64`

Returns total detected usable physical RAM in bytes.

`PhysicalMemoryManager::allocated_memory(&self) -> u64`

Returns total allocated physical RAM in bytes.

`PhysicalMemoryManager::free_memory(&self) -> u64`

Returns currently unallocated physical RAM in bytes.

`x86_64::structures::paging::FrameAllocator<Size4KiB>` Trait Implementation

```
unsafe impl FrameAllocator<Size4KiB> for PhysicalMemoryManager {
    fn allocate_frame(&mut self) -> Option<PhysFrame<Size4KiB>> {
        self.allocate_frame()
    }
}
```

Enables `PhysicalMemoryManager` to serve directly as the frame supplier for `x86_64` page table manipulators and memory allocators.

---

5. UEFI Firmware Integration & Low Memory Protection

```
pub fn init_memory_from_uefi()
```

Queries `uefi::boot::memory_map(MemoryType::LOADER_DATA)`: 1. Iterates through all memory descriptors provided by the UEFI firmware. 2. Filters specifically for `uefi::boot::MemoryType::CONVENTIONAL`. 3. **Reserves the First 1 MiB (`0x100000`)**: Skips addresses below 1 MiB to protect legacy Real-Mode Interrupt Vector Tables (IVT), BIOS Data Areas (BDA), and to prevent `PhysAddr(0)` from being issued as a valid frame (protecting against null-pointer aliasing). 4. Registers each usable conventional memory block into the global `FRAME_ALLOCATOR` mutex.

---

6. Practical Example

```
use crate::memory::frame_allocator::{FRAME_ALLOCATOR, init_memory_from_uefi};

// 1. Read UEFI firmware memory map and discover usable RAM
init_memory_from_uefi();

// 2. Allocate a 4 KiB physical frame
let frame = {
    let mut alloc = FRAME_ALLOCATOR.lock();
    alloc.allocate_frame().expect("System has available RAM")
};

// 3. Inspect frame address
let physical_address = frame.start_address().as_u64();
```

STAGE 4: MEMORY SUBSYSTEM • BOOT SEQUENCE ORDER #18

Dynamic Kernel Heap Allocator Documentation

- **Source File**: `src/memory/heap.rs`
- **Subsystem**: Dynamic Memory Management & Global Allocator (`alloc` crate)

---

1. Conceptual Overview

Operating system kernels initially boot in an environment where standard dynamic allocation (`malloc` or `new`) is unavailable. All structures must fit into static arrays or the stack.

The **Dynamic Kernel Heap Allocator** bridges this gap: - It carves out a designated region of physical/virtual RAM (default: **8 MiB**, 2048 contiguous 4 KiB pages). - It binds this region to Rust's standard allocation interface via `#[global_allocator]`. - It enables the Rust `extern crate alloc;` library across the entire Hydra codebase, unlocking idiomatic heap collections: `Vec`, `String`, `Box`, `Arc`, and `BTreeMap`.

---

2. Allocation Architecture

```
graph TD
    A[Hydra Kernel / Subsystems] ⟶|alloc::vec::Vec / Box / String| B[Rust Core Allocator API]
    B ⟶|GlobalAlloc trait| C[linked_list_allocator::LockedHeap]
    C ⟶|init heap pointer| D[src/memory/heap.rs]
    D ⟶|Step 1: Try Contiguous Conventional RAM| E[src/memory/frame_allocator.rs]
    D ⟶|Step 2: Fallback| F[uefi::boot::allocate_pages LOADER_DATA]
```

Allocation Pipeline:

1. **Physical Backing Allocation**: - `heap::init()` requests a contiguous block of 2,048 physical 4 KiB frames from `FRAME_ALLOCATOR.lock().allocate_contiguous_frames( ... )`. - If not yet ready, it falls back to UEFI firmware `allocate_pages`.
2. **Zero-Initialization**: - Memory is zero-cleared to avoid dirty residual data.
3. **LinkedList Allocator Hookup**: - `HEAP_ALLOCATOR.lock().init(start_ptr, HEAP_SIZE)` configures the free-list structures.
4. **Self-Test Verification**: - `heap::self_test()` validates `Box` allocation, multiple `Vec` capacity expansions, `alloc::format!` string formatting, and checks that memory is fully reclaimed upon RAII drops.

---

3. Public API Breakdown

Function	Signature	Description
`init()`	`fn() → Result<HeapStats, &'static str>`	Initializes the global heap allocator using contiguous physical memory.
`is_initialized()`	`fn() → bool`	Checks if dynamic allocations are currently safe to execute.
`stats()`	`fn() → HeapStats`	Queries current heap metrics (start address, total bytes, used bytes, free bytes).
`self_test()`	`fn() → Result<(), &'static str>`	Executes regression tests validating `Box`, `Vec`, `String`, and RAII deallocation without leaks.

---

4. Interactive Shell Inspection

Hydra provides real-time shell commands to monitor and test the heap:

```
hydra> mem
Physical Memory Manager Statistics:
  Total Usable RAM:    ...
Dynamic Kernel Heap Allocator:
  Heap Status:         ONLINE
  Start Address:       0x...
  Total Heap Size:     8388608 bytes (8 MiB)
  Used Heap Memory:    ...
  Free Heap Memory:    ...

hydra> heap stats
Dynamic Kernel Heap Statistics:
  Status:              ONLINE
  Base Physical/Virt: 0x...
  Total Size:          8388608 bytes (8 MiB)
  Heap Utilization:    0%

hydra> heap test
[*] Running dynamic heap allocation self-test suite...
[+] Heap self-test PASSED:
    - Box<u64> heap allocation & dereference verified
    - Vec<usize> dynamic buffer reallocation (1,000 items) verified
    - alloc::format! dynamic string verified
    - RAII drop cleanup verified without memory leaks
```

STAGE 5: VIRTUAL MEMORY • BOOT SEQUENCE ORDER #19

Virtual Memory & Paging Documentation

- **Source File**: `src/memory/paging.rs`
- **Subsystem**: Memory Management (Virtual Layer & Hardware MMU)

---

1. Conceptual Overview

Modern 64-bit CPUs do not let user programs or high-level kernel functions read and write directly to physical RAM chips. Instead, they use **Virtual Memory**.

Every address a program interacts with is a **Virtual Address**. The hardware Memory Management Unit (MMU) consults a multi-level tree of translation tables (called **Page Tables**) to convert that virtual address into a **Physical Address** in RAM.

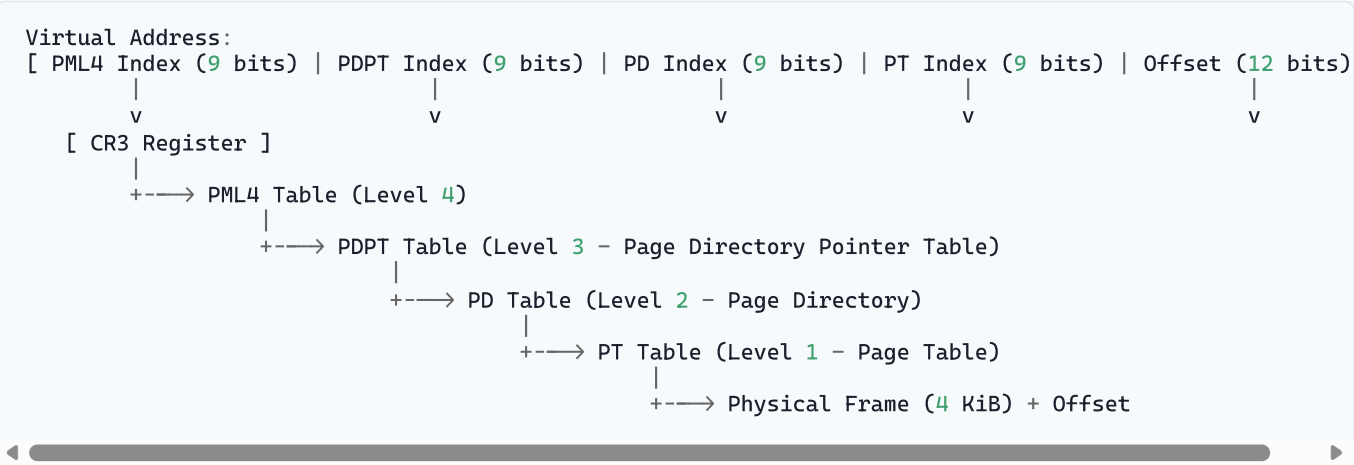
Benefits of Paging

1. **Isolation**: Process A and Process B can both have code at virtual address `0x400000`, but they map to completely different physical RAM chips.
2. **Protection**: Pages can be marked read-only, non-executable (NX/XD bit to prevent malware buffer-overflow attacks), or kernel-only.
3. **Continuity**: Large buffers appear contiguous in virtual memory even if physical frames are scattered across RAM.

---

2. 4-Level Paging Architecture (x86_64)

In 64-bit x86 architecture, translating a 48-bit canonical virtual address uses a 4-level tree:



- **CR3**: The CPU control register holding the physical address of the active Level 4 (PML4) table.

---

3. The `AddressSpace` Struct

```
pub struct AddressSpace {
    pml4_phys: PhysAddr,
}
```

Represents a virtual memory context anchored by its PML4 root table.

---

4. API Reference & Methods

Method	Signature	Description
`active()`	`fn() -> Self`	Attaches to the active address space by reading `CR3`.
`from_pml4()`	`const fn(PhysAddr) -> Self`	Creates an address space handle for a given root PML4 table.
`switch()`	`unsafe fn(&self)`	Reloads the CPU's `CR3` register with this address space's PML4 root.
`translate_addr()`	`unsafe fn(&self, VirtAddr) -> Option<PhysAddr>`	Walks PML4 hierarchy to translate any virtual address to its physical address.
`translate_page()`	`unsafe fn(&self, Page) -> Option<(PhysFrame, PageTableFlags)>`	Queries mapped frame and protection flags (`PRESENT`, `WRITABLE`, `USER`, `NX`).
`map_page()`	`unsafe fn(&mut self, Page, PhysFrame, PageTableFlags, &mut PhysicalMemoryManager) -> Result<(), &'static str>`	Allocates missing intermediate tables (PDPT, PD, PT) on demand, maps a 4 KiB page, and flushes the TLB.
`unmap_page()`	`unsafe fn(&mut self, Page) -> Result<PhysFrame, &'static str>`	Clears the page table entry, flushes the TLB via `invlpg`, and returns the previously mapped frame.
`update_flags()`	`unsafe fn(&mut self, Page, PageTableFlags) -> Result<(), &'static str>`	Updates protection bits for an existing mapped page and invalidates TLB.
`identity_map()`	`unsafe fn(&mut self, PhysFrame, PageTableFlags, &mut PhysicalMemoryManager) -> Result<(), &'static str>`	Maps a physical frame 1:1 (`virt = phys`), essential for MMIO (APIC, GOP framebuffer).
`map_range()`	`unsafe fn(&mut self, Page, PhysFrame, usize, PageTableFlags, &mut PhysicalMemoryManager) -> Result<(), &'static str>`	Maps a contiguous sequence of virtual pages to physical frames.
`create_user_space()`	`fn(&mut PhysicalMemoryManager) -> Result<Self, &'static str>`	Allocates a new PML4 table, mirrors higher-half kernel entries, and leaves lower-half empty for isolated Ring 3 user tasks.

5. Interactive Shell Commands

Hydra provides built-in commands to inspect and modify virtual memory mappings dynamically:

```
hydra> translate 0x400000
4-Level PML4 Page Table Walk for 0x400000:
  PML4 Index (L4):    0
  PDPT Index (L3):    0
  PD Index   (L2):    2
  PT Index   (L1):    0
  Page Offset:        0x0
  Status:             MAPPED
  Physical Target:    0x400000
  Base Frame:         0x400000
  Protection Flags:   [PRESENT] [WRITABLE] [SUPERVISOR/RING-0] [EXECUTABLE]

hydra> map 0x7000000000 0x1000000 rw
[+] Successfully mapped Page 0x7000000000 → Frame 0x1000000 (flags: PageTableFlags(PRESENT | WRITABLE

hydra> unmap 0x7000000000
[+] Successfully unmapped Page 0x7000000000 (previously mapped to Frame 0x1000000)
```

STAGE 6: SECURITY & CAPABILITIES • BOOT SEQUENCE ORDER #20

Capability Subsystem Overview

- **Source File**: `src/capability/mod.rs`
- **Subsystem**: Capability-Based Security

---

1. Conceptual Overview

Traditional operating systems (like standard Linux or Windows) often rely on **Access Control Lists (ACLs)** or ambient authority (e.g., "this process runs as root/administrator, so it can touch anything"). If an administrative process has a bug or is hijacked, the entire system is compromised.

Hydra uses Capability-Based Security (similar to seL4 and Fuchsia Zircon).

The Real-World Analogy

Think of a **Capability** as a non-forgeable, digital smart keycard: - You cannot just claim "I am the administrator". - You can only perform an operation if you hold a specific keycard that explicitly grants the right to perform that action on that exact resource. - You can hand a copy of a keycard with restricted permissions to another process (e.g., giving someone a key that allows reading a file, but not modifying or deleting it).

---

2. The `Rights` Bitflags

`Rights` defines what actions a process holding a capability is permitted to execute:

```
#[derive(Debug, Clone, Copy, PartialEq, Eq)]
pub struct Rights(pub u32);
```

Supported Rights Flags

Flag	Value	Description
`Rights::READ`	`1 << 0` (1)	Permission to inspect data in the target object.
`Rights::WRITE`	`1 << 1` (2)	Permission to modify or overwrite the target object.
`Rights::EXECUTE`	`1 << 2` (4)	Permission to execute machine instructions from the object (memory frame).
`Rights::GRANT`	`1 << 3` (8)	Permission to delegate, duplicate, or transfer this capability to another process.
`Rights::CALL`	`1 << 4` (16)	Permission to invoke or call an IPC Endpoint.
`Rights::ALL`	`0x1F` (31)	Full combined permissions (`READ | WRITE | EXECUTE | GRANT | CALL`).

Checking Permissions

- `rights.contains(required)` : Returns `true` if all bits in `required` are present.

---

3. The `CapType` Enum

`CapType` defines what kind of physical or virtual kernel resource the capability represents:

```
#[derive(Debug, Clone, Copy, PartialEq, Eq)]
pub enum CapType {
    UntypedMemory { size_bytes: usize },
    MemoryFrame { physical_addr: u64 },
    AddressSpace { root_cr3: u64 },
    Endpoint { endpoint_id: u32 },
    Thread { thread_id: u32 },
    HardwareDevice { device_id: u32 },
}
```

- `UntypedMemory` : Raw, unallocated physical memory blocks that can be carved up into specific objects.
- `MemoryFrame` : A specific 4096-byte page of physical RAM.
- `AddressSpace` : A virtual memory context (represented by the x86_64 CR3 root page table).
- `Endpoint` : A communication mailbox for IPC.
- `Thread` : A schedulable execution thread.
- `HardwareDevice` : An I/O hardware device (such as a GPU, network card, or timer).

---

4. The `Capability` Struct

```
#[derive(Debug, Clone, Copy)]
pub struct Capability {
    pub cap_type: CapType,
    pub rights: Rights,
}
```

Combines the resource descriptor (`cap_type`) with the allowable permissions (`rights`).

Constructor

```
let memory_cap = Capability::new(
    CapType::MemoryFrame { physical_addr: 0x1000 },
    Rights(Rights::READ.0 | Rights::WRITE.0)
);
```

STAGE 6: SECURITY & CAPABILITIES • BOOT SEQUENCE ORDER #21

Capability Table Module Documentation

- **Source File**: `src/capability/table.rs`
- **Subsystem**: Capability-Based Security

---

1. Conceptual Overview

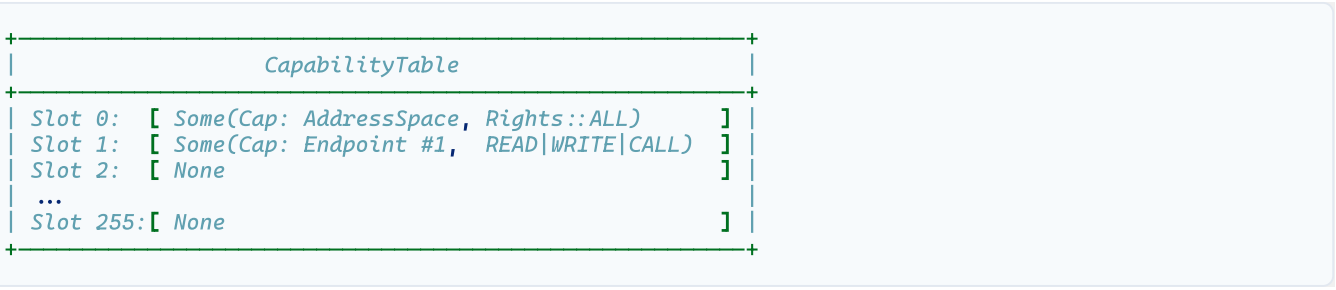
In Hydra, userland processes cannot store raw pointers to kernel objects or hardware devices. Instead, the kernel maintains a **Capability Table** (also known as a **CSpace** or **CNode**) on behalf of each process.

The Real-World Analogy

Think of a **Capability Table** like a keyring or safe-deposit locker: - The process holds numbers (slots: `0`, `1`, `2`, ...). - When a process asks the kernel, *"Read from memory at Slot 3"*, the kernel checks Slot 3 in that process's Capability Table. - If Slot 3 contains a valid capability and includes `Rights::READ`, the kernel performs the operation. - If the slot is empty or lacks the required right, the kernel immediately denies the request. - The process never touches the raw physical address or hardware pointer directly, preventing spoofing and unauthorized access.

---

2. Architecture & Constants



- `MAX_CAPABILITIES: usize = 256`: Each table can hold up to 256 active capabilities per task in fixed array storage, avoiding dynamic heap allocation.

---

3. The `CapabilityTable` Struct

```rust
pub struct CapabilityTable {
    slots: [Option<Capability>; MAX_CAPABILITIES],
    next_free_slot: usize,
}
```

- `slots`: Fixed array of 256 optional capability slots.
- `next_free_slot`: Circular search cursor to quickly find available slots upon insertion.

---

4. API Reference & Methods

`CapabilityTable::new() → Self`

Creates a new, empty capability table where all 256 slots are `None`.

`CapabilityTable::insert(&mut self, cap: Capability) → Result<usize, &'static str>`

Allocates the first available empty slot, stores the capability, and returns the slot index (`0..255`). - **Returns** `Ok(slot_index)` on success. - **Returns** `Err("Capability table full")` if all 256 slots are currently occupied.

`CapabilityTable::get(&self, slot: usize) → Option<&Capability>`

Retrieves a read-only reference to the capability stored at the specified slot. - Returns `None` if the slot is out of bounds or empty.

`CapabilityTable::revoke(&mut self, slot: usize) → Option<Capability>`

Removes and clears the capability at `slot`, effectively revoking the process's access to that resource.

`CapabilityTable::check_rights(&self, slot: usize, required_rights: Rights) → bool`

Helper method that verifies: 1. Whether `slot` contains an active capability. 2. Whether the capability possesses all bits defined in `required_rights`.

---

5. Practical Example

```
use crate::capability::table::CapabilityTable;
use crate::capability::{Capability, CapType, Rights};

let mut cspace = CapabilityTable::new();

// 1. Grant access to a hardware memory frame
let frame_cap = Capability::new(
    CapType::MemoryFrame { physical_addr: 0x2000_0000 },
    Rights(Rights::READ.0 | Rights::WRITE.0)
);

// 2. Insert into the process table
let slot = cspace.insert(frame_cap).expect("Table has free slots");

// 3. Verify permissions before performing an operation
if cspace.check_rights(slot, Rights::WRITE) {
    // Permission granted: perform write
} else {
    // Permission denied: trigger error
}
```

STAGE 7: IPC SUBSYSTEM • BOOT SEQUENCE ORDER #22

IPC Subsystem Overview

- **Source File**: `src/ipc/mod.rs`
- **Subsystem**: Microkernel IPC (Inter-Process Communication)

---

1. Role in Hydra OS

The `ipc` module binds together the fundamental primitives for Inter-Process Communication in the Hydra microkernel.

Because Hydra isolates all system services (network stack, file systems, device drivers, cloud node daemons) into separate userland processes, fast and secure message passing is the backbone of the entire operating system.

---

2. Module Organization

- `endpoint.rs`: Provides the `Endpoint` and `SharedEndpoint` message queues.
- `message.rs`: Defines the structure of an IPC `Message`, including inline arguments and security capability tokens.

---

3. Re-exported Types

```
pub use endpoint::Endpoint;
pub use message::Message;
```

These re-exports allow any other module in the kernel to conveniently import IPC types directly via `use crate::ipc::{Endpoint, Message};`.

STAGE 7: IPC SUBSYSTEM • BOOT SEQUENCE ORDER #23

IPC Message Module Documentation

- **Source File**: `src/ipc/message.rs`
- **Subsystem**: Microkernel IPC (Inter-Process Communication)

---

1. Conceptual Overview

In a microkernel operating system like **Hydra**, core operating system services (such as file systems, network drivers, and device managers) run as separate, isolated programs rather than inside a single monolithic kernel. Because these isolated programs cannot directly access each other's memory, they must communicate by sending **Messages** across a secured boundary.

The Real-World Analogy

Think of an **IPC Message** like a registered letter or courier package: 1. **The Label (`label`)**: The subject line or action request (e.g., "Read File", "Get Time", "Allocate Memory"). 2. **The Fast Payload (`args`)**: A quick handwritten note with up to 4 numbers or pointers that fit directly into the CPU's registers for high-speed delivery without touching RAM. 3. **The Enclosed Key/Passport (`capability`)**: An optional security token granted to the receiver, giving them special permission to access a specific memory buffer, device, or channel.

---

2. Architecture & Data Structures

```
+-------------------------------------------------------------+
|                          Message                            |
+-------------------------------------------------------------+
| label: u32                | Protocol / Syscall ID           |
| args: [u64; 4]            | Up to 4 fast 64-bit parameters  |
| capability: Option<Cap>   | Optional transferred security   |
|                           | token / permission              |
+-------------------------------------------------------------+
```

Constants

- `MAX_INLINE_WORDS: usize = 4` : The number of 64-bit arguments (`u64`) carried directly within a message. Using 4 words (32 bytes) allows the microkernel to transfer arguments directly using CPU registers (such as `RDI` , `RSI` , `RDX` , `RCX` on x86_64), making IPC fast.

---

3. The `Message` Struct

```rust
#[derive(Debug, Clone, Copy)]
pub struct Message {
    pub label: u32,
    pub args: [u64; MAX_INLINE_WORDS],
    pub capability: Option<Capability>,
}
```

Fields Explained

Field	Type	Description
`label`	`u32`	An operation code indicating what action is requested (e.g. `0x1001` for ping, `0x2001` for read).
`args`	`[u64; 4]`	Array of 4 inline 64-bit numbers. Can be integers, memory offsets, or byte lengths.
`capability`	`Option<Capability>`	An optional security token transferred along with the message to delegate authority.

---

4. API Reference & Methods

`Message::new(label: u32) → Self`

Creates a new message with the given operation label. All 4 arguments are initialized to `0`, and no capability is attached.

```rust
let msg = Message::new(0x01); // Simple ping message
```

`Message::with_args(label: u32, args: [u64; MAX_INLINE_WORDS]) → Self`

Creates a new message with an operation label and up to 4 inline arguments.

```rust
// Example: requesting to read 512 bytes starting at offset 1024
let msg = Message::with_args(0x1002, [1024, 512, 0, 0]);
```

`Message::with_cap(mut self, cap: Capability) → Self`

Builder pattern method to attach a capability to an existing message. This transfers access rights to the receiving process.

```rust
let msg = Message::new(0x2000).with_cap(shared_memory_cap);
```

---

5. Practical Example

```rust
use crate::capability::{Capability, CapType, Rights};
use crate::ipc::Message;

// 1. Create a capability granting READ access to an endpoint
let ep_cap = Capability::new(CapType::Endpoint { endpoint_id: 10 }, Rights::READ);

// 2. Prepare an IPC message carrying data and delegating the capability
let message = Message::with_args(0x1001, [42, 1337, 0, 0])
    .with_cap(ep_cap);

// 3. The message is now ready to be sent over an Endpoint!
```

STAGE 7: IPC SUBSYSTEM • BOOT SEQUENCE ORDER #24

IPC Endpoint Module Documentation

- **Source File**: `src/ipc/endpoint.rs`
- **Subsystem**: Microkernel IPC (Inter-Process Communication)

---

1. Conceptual Overview

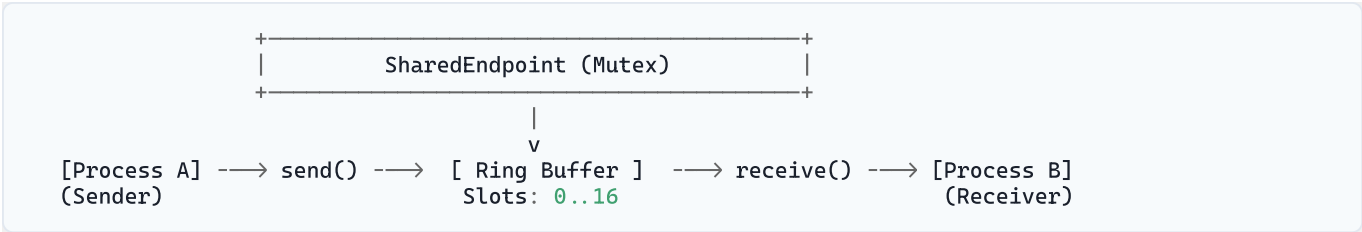
In a microkernel architecture, programs cannot talk directly to each other because their address spaces are completely isolated by the CPU's memory management unit (MMU). An **Endpoint** acts as a meeting place (rendezvous point) or postal mailbox managed by the kernel.

The Real-World Analogy

Imagine an apartment building where residents cannot enter each other's apartments: - An **Endpoint** is a secure PO Box in the lobby. - Process A drops a `Message` into the box (`send`). - Process B checks the box and retrieves the message (`receive`). - The kernel ensures only residents holding the correct security key (`Capability`) can post or retrieve items from that specific box.

---

2. Architecture & Data Structures



Queue Parameters

- `QUEUE_CAPACITY: usize = 16` : The maximum number of unread messages the endpoint buffer can hold before rejecting new incoming messages with an error.

---

3. Struct Details

`Endpoint`

The core internal endpoint managing a circular ring buffer of messages.

```
pub struct Endpoint {
    id: u32,
    buffer: [Option<Message>; QUEUE_CAPACITY],
    head: usize,
    tail: usize,
    count: usize,
}
```

- `id`: A unique numerical identifier for this endpoint.
- `buffer`: Fixed-size array holding pending messages without needing dynamic heap allocations.
- `head`: Read pointer pointing to the next message to be consumed (FIFO order).
- `tail`: Write pointer pointing to the next free slot for an incoming message.
- `count`: Current number of queued, unread messages.

`EndpointState`

Indicates the scheduling and readiness state of the communication channel:

```
#[derive(Debug, Clone, Copy, PartialEq, Eq)]
pub enum EndpointState {
    Idle,
    Active,
}
```

`SharedEndpoint`

A thread-safe wrapper around `Endpoint` using a `spin::Mutex`. This allows concurrent threads and interrupts to safely send and receive messages without race conditions.

```
pub struct SharedEndpoint(Mutex<Endpoint>);
```

---

4. API Reference & Methods

`Endpoint::new(id: u32) -> Self`

Initializes a new empty message queue with the specified ID.

`Endpoint::send(&mut self, msg: Message) -> Result<(), &'static str>`

Pushes a message into the queue: - **Returns** `Ok(())` if the message was placed into the buffer. - **Returns** `Err("Endpoint message queue is full")` if all 16 slots are occupied.

`Endpoint::receive(&mut self) -> Option<Message>`

Pulls the oldest message out of the queue (FIFO): - **Returns** `Some(Message)` if a message is available. - **Returns** `None` if the queue is currently empty.

`Endpoint::pending_count(&self) -> usize`

Returns how many messages are waiting to be read.

`SharedEndpoint::new(id: u32) → Self`

Constructs a new mutex-protected endpoint.

`SharedEndpoint::send(&self, msg: Message) → Result<(), &'static str>`

Acquires the spinlock and sends a message into the underlying endpoint.

`SharedEndpoint::receive(&self) → Option<Message>`

Acquires the spinlock and consumes the next available message.

`SharedEndpoint::id(&self) → u32`

Returns the ID of the wrapped endpoint.

---

5. Practical Example

```
use crate::ipc::{Endpoint, Message};

// 1. Create a communication endpoint with ID 1
let mut mailbox = Endpoint::new(1);

// 2. Sender prepares and sends a message
let request = Message::with_args(0x100, [1, 2, 3, 4]);
mailbox.send(request).expect("Queue has space");

// 3. Check queue status
assert_eq!(mailbox.pending_count(), 1);

// 4. Receiver reads the message
if let Some(msg) = mailbox.receive() {
    // Process msg.label and msg.args ...
}
```

STAGE 8: FILE SERVER & STORAGE • BOOT SEQUENCE ORDER #25

ZFS Microkernel File Server Documentation

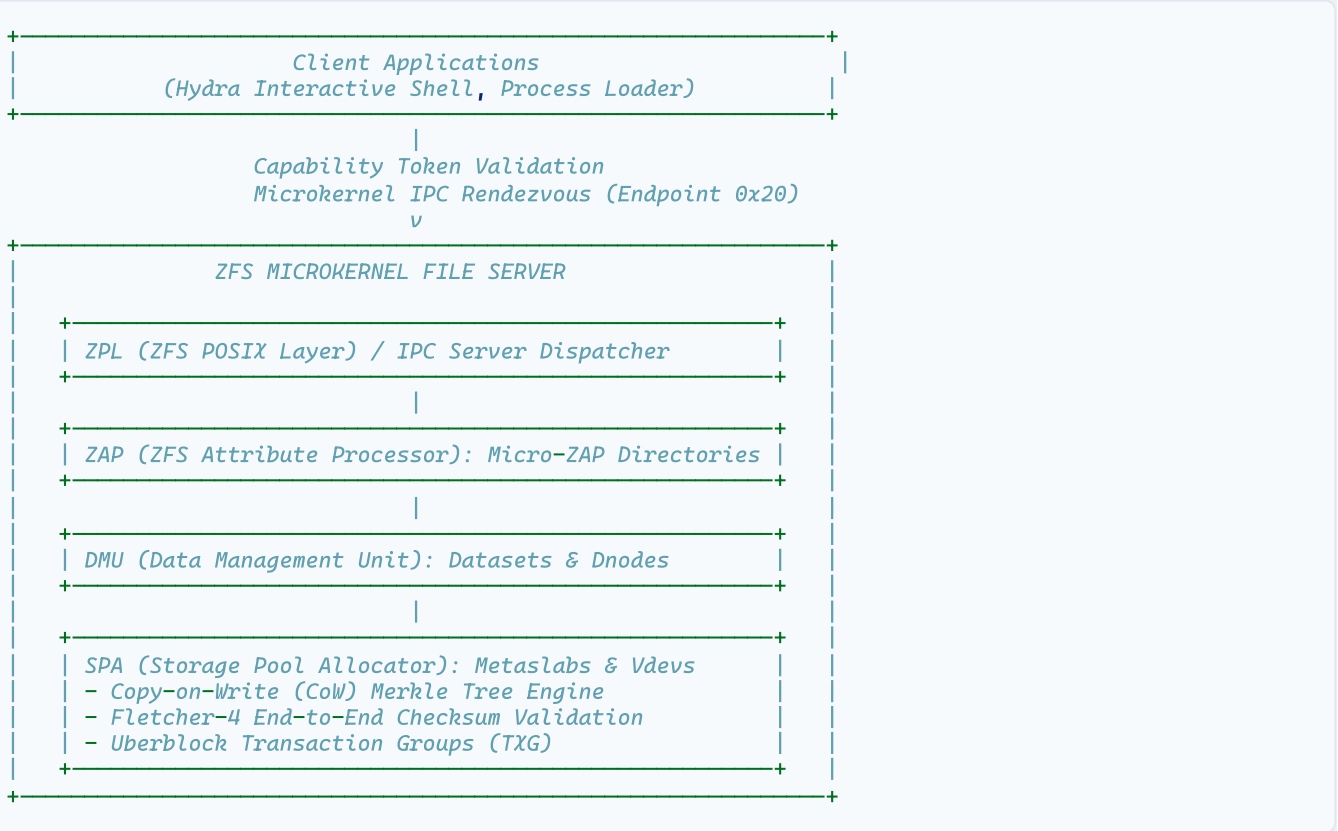
- **Source Directory**: `src/fs/zfs/` & `src/fs/client.rs`
- **Subsystem**: Default Filesystem Architecture & Standalone File Server
- **Endpoint**: `0x20` (`ENDPOINT_ZFS_SERVER`)

---

1. Architectural Motivation

In traditional monolithic operating systems, the filesystem runs in kernel space (Ring 0) with direct block device access. This violates principle-of-least-privilege and microkernel design tenants. A single bug or corruption in the filesystem driver can compromise the entire operating system kernel.

Hydra implements **ZFS (Zettabyte File System)** as the default filesystem, structured strictly as an **autonomous Microkernel File Server (`ZfsServer`)**.



---

2. ZFS Core Architecture

Hydra's ZFS implementation faithfully models the fundamental layers of OpenZFS:

1. Storage Pool Allocator (SPA)

- **Storage Pool**: Named `hydrapool` by default.
- **VDEV (Virtual Device)**: Manages physical block storage or memory-backed frames.
- **Copy-on-Write (CoW)**: Data blocks are never updated in-place. Writing allocates new blocks, preserves previous data for snapshots, and links to a new Merkle root.
- **Active Uberblock**: The root of the storage pool (`magic: 0x00bab10c` , `version: 5000`). Every write transaction commits an incremented Transaction Group (`TXG`).

2. Fletcher-4 End-to-End Checksums

Every block pointer (`blkptr_t`) carries a 256-bit Fletcher-4 checksum (4 parallel 64-bit accumulators): $$\text{Fletcher-4: } a = \sum w_i, \quad b = \sum a_i, \quad c = \sum b_i, \quad d = \sum c_i$$ Upon read operations, the ZFS File Server recalculates the checksum and detects any silent data corruption before returning data to the caller.

3. Data Management Unit (DMU) & Dnodes

- Objects in ZFS are represented by `DnodePhys` structures (`obj_type` , block size, file size, direct block pointers).
- Supported Object Types:
- `PlainFile` : Data payload blocks.
- `Directory` : Associated with a ZAP object.
- `Dataset` : Isolated filesystem namespace (`hydrapool/root` , `hydrapool/bin`).
- `MasterNode` : Meta-Object Set (MOS) root.

4. ZFS Attribute Processor (ZAP)

- Micro-ZAP structure indexing directory entries (mapping string filenames to object IDs).
- Enables $O(1)$ file lookups inside datasets.

---

3. Microkernel IPC Protocol (`Endpoint #0x20`)

Clients interact with the File Server by sending `Message` envelopes through `ENDPOINT_ZFS_SERVER` .

Protocol Label	Opcode	Arguments (`args`)	Return Values	Description
`ZFS_IPC_PING`	`0x3001`	-	`[status, txg, 0, 0]`	Server health check and active TXG.
`ZFS_IPC_LOOKUP`	`0x3002`	`[dir_obj, name_hash, 0, 0]`	`[status, obj_id, obj_type, 0]`	Resolves filename to object ID.
`ZFS_IPC_READ`	`0x3003`	`[obj_id, dest_buf_ptr, max_len, 0]`	`[status, bytes_read, txg, 0]`	Reads and checksum-verifies file data.
`ZFS_IPC_WRITE`	`0x3004`	`[obj_id, src_buf_ptr, len, 0]`	`[status, new_txg, 0, 0]`	CoW writes data and commits new TXG.
`ZFS_IPC_POOL_STATUS`	`0x3006`	-	`[status, total_mb, alloc_kb, free_txg]`	Reports pool metrics and health.
`ZFS_IPC_SNAPSHOT`	`0x3007`	`[dataset_id, 0, 0, 0]`	`[status, txg, 0, 0]`	Creates point-in-time snapshot.

---

4. Capability-Based Security Enforcement

The File Server verifies client capabilities on every request: - **Read Requests**: The client must present a capability containing `Rights::READ`. If missing, the server responds with `ZFS_STATUS_PERMISSION_DENIED`. - **Write / CoW Transactions**: Requires `Rights::WRITE`. - **Snapshots**: Requires `Rights::WRITE`.

---

5. Interactive Shell Commands

Hydra provides native user commands to interact with the ZFS File Server:

```
hydra> zpool status
  pool: hydrapool
 state: ONLINE
status: The pool is operating normally via the Microkernel ZFS File Server.
action: No known data errors detected; all blocks checksum-verified.
config:
        NAME                STATE     READ WRITE CKSUM
        hydrapool           ONLINE       0     0     0
          vdev-ramdisk-0    ONLINE       0     0     0

Pool Properties & Metrics:
  - Capacity:           8 MiB
  - Allocated:          128 KiB
  - Free Space:         8064 KiB
  - Active Uberblock:   TXG #2
  - Checksum Algorithm: Fletcher-4 (4x 64-bit words)
  - Allocation Policy:  Copy-on-Write (CoW) Merkle Tree

hydra> zfs list
NAME                    USED    AVAIL MOUNTPOINT
-------------------------------------------------------
hydrapool/root          128K    8064K /
hydrapool/bin           128K    8064K /bin

hydra> zfs cat readme.txt
Hydra OS Default Filesystem: OpenZFS Architecture
Integrated as an independent Microkernel File Server.
All operations are verified via Fletcher-4 checksums!

hydra> zfs write demo.txt "Hello from ZFS on Hydra Microkernel!"
[+] CoW write committed to ZFS File Server!
    - File: 'demo.txt' (36 bytes)
    - Merkle Root Updated to TXG #3
    - Fletcher-4 Checksum computed & stored
```

STAGE 8: FILE SERVER & STORAGE • BOOT SEQUENCE ORDER #26

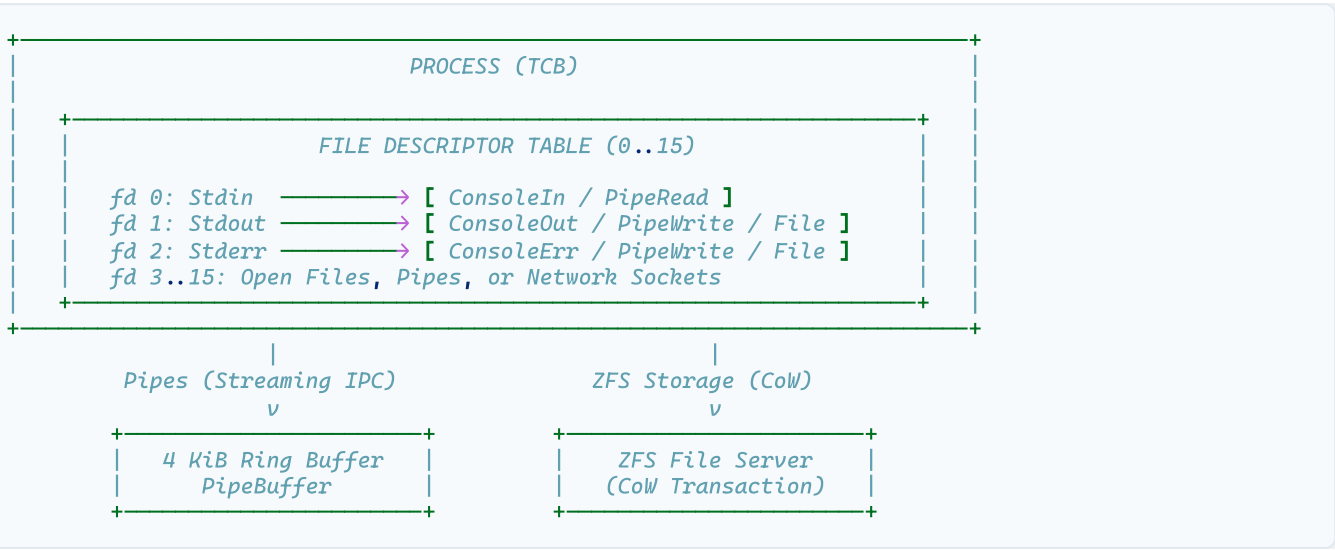
Unix File Descriptor Table & Shell Pipelines

- **Source Files**: `src/fs/fd.rs`, `src/fs/pipe.rs`, `src/shell/pipeline.rs`
- **Subsystem**: Virtual File System (VFS), POSIX File Descriptors, IPC Pipes, Shell Streams
- **Privilege Levels**: Ring 0 (Kernel Subsystem) & Ring 3 (Userland Syscalls)

---

1. Architectural Overview

Hydra OS implements a complete Unix-like File Descriptor model that decouples input/output operations from underlying hardware devices or filesystem implementations. Every process in Hydra maintains its own **File Descriptor Table (`FdTable`)**, allowing standard streams to be seamlessly redirected across processes, anonymous pipes, and ZFS files.



---

2. Standard Streams & Descriptors

Every newly spawned task automatically inherits a standard Unix descriptor table:

FD Index	POSIX Name	Default Target	Capability & Access
`0`	`STDIN_FILENO`	`FileDescriptor::ConsoleIn`	PS/2 Keyboard Input Buffer (Non-blocking)
`1`	`STDOUT_FILENO`	`FileDescriptor::ConsoleOut`	Physical Framebuffer / Terminal Screen
`2`	`STDERR_FILENO`	`FileDescriptor::ConsoleErr`	Physical Framebuffer / Error Channel
`3..15`	Available FDs	`None`	Allocated dynamically via `open()` or `pipe()`

File Descriptor Types (`FileDescriptor`)

1. `ConsoleIn` : Reads characters from the hardware PS/2 keyboard ring buffer.
2. `ConsoleOut` / `ConsoleErr` : Writes formatted ASCII/UTF-8 directly to the screen console.
3. `PipeRead(Arc<Mutex<PipeBuffer>>)` : Reads streamed bytes from an anonymous pipe ring buffer.
4. `PipeWrite(Arc<Mutex<PipeBuffer>>)` : Writes streamed bytes into an anonymous pipe ring buffer.
5. `File` : Represents a stateful cursor into a file stored on the ZFS storage pool.

---

3. Anonymous Pipe Subsystem

The `PipeBuffer` primitive (`src/fs/pipe.rs`) provides high-throughput, lock-free/spinlock-protected unidirectional byte streaming between processes:

- **Buffer Size**: 4096 bytes circular ring buffer.
- **Reference Counting**: Tracks active reader and writer file descriptors.
- **EOF Semantics**: When all writers close their end (`writers == 0`), subsequent reads on an empty buffer return `0` bytes (standard POSIX EOF).
- **Broken Pipe (EPIPE)**: Attempting to write to a pipe with `readers == 0` immediately fails with `-EPIPE` (or broken pipe error).

---

4. Unix System Calls

Hydra OS exposes the full complement of POSIX I/O system calls to Ring 3 processes:

Syscall	Number	Arguments	Description
`SYS_WRITE`	`3`	`arg0: fd` , `arg1: ptr` , `arg2: count`	Writes bytes to the specified file descriptor
`SYS_OPEN`	`9`	`arg0: path_ptr` , `arg1: len` , `arg2: flags`	Opens or creates a file in ZFS (`O_RDONLY` , `O_WRONLY` , `O_RDWR` , `O_APPEND`)
`SYS_CLOSE`	`10`	`arg0: fd`	Closes the file descriptor and decrements pipe refcounts
`SYS_READ`	`11`	`arg0: fd` , `arg1: buf_ptr` , `arg2: count`	Reads bytes from a descriptor (Console, Pipe, or File)
`SYS_PIPE`	`12`	`arg0: *mut [u64; 2]`	Allocates a new pipe and returns `[read_fd, write_fd]`
`SYS_DUP2`	`13`	`arg0: oldfd` , `arg1: newfd`	Duplicates `oldfd` onto `newfd` , redirecting streams

---

5. Shell Pipelines & Redirection

The Hydra Interactive Shell (`src/shell/pipeline.rs`) supports arbitrary pipeline chaining and file redirection:

1. Pipelines (`|`)

Chains the standard output (`fd 1`) of command N to the standard input (`fd 0`) of command $N+1$:

```
cat /test.txt | grep Pipeline
ps | grep idle
help | grep fs
```

2. Output Redirection (`>` and `>>`)

Redirects command standard output to a file in ZFS: - `>` (Overwrite / Truncate): `bash echo Hello Hydra > /test.txt` - `>>` (Append): `bash echo Pipeline Verified >> /test.txt`

3. Combined Pipeline & Redirection

Pipelines can seamlessly terminate in a file redirection target:

```
ps | grep idle > /idle_process.txt
```

Output is buffered in memory during pipeline execution and committed atomically to the ZFS filesystem via Copy-on-Write (CoW).

STAGE 8B: HARDWARE VIDEO & FRAMEBUFFER • BOOT SEQUENCE ORDER #27

Hardware Video Driver & GOP Framebuffer Documentation

- **Source Module**: `src/video/`
- **Subsystem**: Display Driver, Linear Framebuffer & Autonomous Video Console

---

1. Conceptual Overview

Before an operating system can terminate firmware boot services (`ExitBootServices`), it must take complete ownership of the hardware display. Otherwise, text and graphics output will terminate with the firmware.

Hydra's video subsystem: 1. Discovers the **UEFI Graphics Output Protocol (GOP)** while boot services are active. 2. Retrieves the hardware's 32-bit linear physical framebuffer pointer, resolution (e.g. 1024x768 or native display), stride, and pixel format (`Rgb8` or `Bgr8`). 3. Uses Hydra's Virtual Memory Manager (`AddressSpace::map_range`) to map the entire physical framebuffer into Hydra's autonomous PML4 page table. 4. Integrates a lightweight **8x16 bitmap font engine** covering all printable ASCII characters. 5. Runs an **autonomous text-mode terminal emulator** directly on the pixel buffer, enabling `print!`, `println!`, colors, line wrapping, and hardware-accelerated vertical scrolling independent of UEFI firmware.

---

2. Video Architecture & Data Flow

```
graph TD
    A[UEFI Firmware POST] ⟶ B[uefi::proto::console::gop::GraphicsOutput]
    B ⟶ C[init_gop: Extract Base PhysAddr, Resolution, Stride, RGB/BGR]
    C ⟶ D[VMM: AddressSpace::map_range Framebuffer into Autonomous PML4]
    D ⟶ E[Framebuffer: Direct Pixel Access, Rect Fill, Glyph Blitting]
    E ⟶ F[VideoConsole: 8x16 Text Matrix, Scrolling, Colors]
    F ⟶ G[Unified Console: stdout + Framebuffer Synchronous Rendering]
    G ⟶ H[ExitBootServices Ready: Display Output Persists Standalone]
```

---

3. Module Breakdown

Source File	Role & Implementation
`src/video/gop.rs`	Hardware discovery via UEFI GOP, physical framebuffer extraction, and VMM page mapping.
`src/video/font.rs`	8x16 ASCII bitmap font table covering characters 32 to 126.
`src/video/framebuffer.rs`	Direct linear pixel buffer controller: `put_pixel`, `fill_rect`, `clear`, `draw_char`, `draw_string`, and `scroll_up` (`core::ptr::copy`).
`src/video/console.rs`	Autonomous text terminal running on the pixel buffer. Implements `core::fmt::Write`, cursor management, line wrapping, and color mapping.
`src/video/mod.rs`	Video subsystem coordinator and public API exports.

---

4. Interactive Shell Inspection & Testing

Hydra provides the `video` (aliases: `gop`, `fb`) command in the shell:

```
hydra> video info
Hardware Video Subsystem & Framebuffer Statistics:
  Driver Status:      ONLINE (Direct GOP Linear Framebuffer)
  Resolution:         1024 x 768 pixels (32 bpp)
  Pixel Stride:       1024 pixels/scanline (4096 bytes)
  Pixel Encoding:     BGR888 (Little-Endian)
  Video Memory Base:  0xe0000000
  Framebuffer Size:   3145728 bytes (3 MiB)
  Text Terminal Grid: 128 columns x 48 rows (8x16 font)
  Kernel VMM Status:  Identity-mapped in Hydra Autonomous PML4

hydra> video test
[+] Graphical test pattern & Wayland preview window rendered to screen.

hydra> video clear
[+] Framebuffer display cleared.
```

STAGE 9: MULTITASKING & SCHEDULER • BOOT SEQUENCE ORDER #28

Task Control Block (TCB) & Thread State

Overview

The **Task Control Block (`Task` / `TCB`)** forms the foundational abstraction of thread execution and process representation in the Hydra 64-bit microkernel. Each execution thread is managed through an independent TCB that encapsulates its unique 64-bit identifier (`TaskId`), descriptive thread label, lifecycle execution state (`TaskState`), scheduling priority (`TaskPriority`), architectural register frame (`TaskRegisters`), dedicated 16 KiB execution stack, CPU tick consumption statistics, and an isolated per-task capability security space (`CapabilityTable`).

The implementation is located in `src/task/tcb.rs` with public re-exports in `src/task/mod.rs`.

Architectural Context Frame (`TaskRegisters`)

When preemptive multitasking switches between tasks, the CPU architectural state must be preserved and restored with bit-exact fidelity. The `TaskRegisters` structure encapsulates all x86_64 general-purpose registers, the instruction pointer (`RIP`), code segment selector (`CS`), processor flags (`RFLAGS`), stack pointer (`RSP`), stack segment selector (`SS`), and the 4-level paging root (`CR3`).

```
#[repr(C)]
#[derive(Debug, Clone, Copy)]
pub struct TaskRegisters {
    pub r15: u64,
    pub r14: u64,
    pub r13: u64,
    pub r12: u64,
    pub rbp: u64,
    pub rbx: u64,
    pub r11: u64,
    pub r10: u64,
    pub r9: u64,
    pub r8: u64,
    pub rdi: u64,
    pub rsi: u64,
    pub rdx: u64,
    pub rcx: u64,
    pub rax: u64,
    pub rip: u64,
    pub cs: u64,
    pub rflags: u64,
    pub rsp: u64,
    pub ss: u64,
    pub cr3: u64,
}
```

Context Synchronization

Two symmetric methods bridge the gap between hardware interrupt frames and the kernel's scheduler data structures:

1. `save_from_context(&mut self, ctx: &ExceptionContext, cr3: u64)`: Captures the running thread's state directly from the stack-saved `ExceptionContext` created by the assembly interrupt gate trampoline, recording the current `CR3` paging root.
2. `restore_to_context(&self, ctx: &mut ExceptionContext)`: Transfers the incoming thread's register frame into the active interrupt context frame on the stack. When the hardware interrupt routine exits via `iretq`, the CPU immediately restores all general-purpose registers and atomically resumes execution inside the new task.

Thread Lifecycle State Machine (`TaskState`)

Each kernel task transitions across five discrete states during its execution lifecycle:

```
stateDiagram-v2
    [*] ⟶ Ready: spawn()
    Ready ⟶ Running: Scheduler timeslice dispatch
    Running ⟶ Ready: Timeslice expired / Preempted (100 Hz PIT)
    Running ⟶ Sleeping: sleep_ms() / sleep_ticks()
    Sleeping ⟶ Ready: Timer tick expiration (PIT ticks ≥ wake_tick)
    Running ⟶ Blocked: Capability IPC wait / rendezvous
    Blocked ⟶ Ready: IPC message available
    Running ⟶ Dead: exit() / return from trampoline
    Ready ⟶ Dead: kill()
    Sleeping ⟶ Dead: kill()
    Blocked ⟶ Dead: kill()
    Dead ⟶ [*]
```

State Definitions

State	Description	Transition Trigger
`Ready`	Thread is runnable and waiting in the run-queue for CPU allocation.	Task creation, timeslice preemption, or sleep expiration.
`Running`	Thread currently owns the CPU core and is executing instructions.	Selected by the scheduler during context switch.
`Sleeping { wake_tick }`	Thread is suspended until the hardware PIT tick counter reaches `wake_tick`.	Invocation of `sleep_ms()` or `sleep_ticks()`.
`Blocked`	Thread is waiting on a capability-mediated IPC message or resource lock.	Synchronous IPC rendezvous request.
`Dead`	Thread has finished execution or was explicitly terminated by PID.	Invocation of `exit()` or administrative `kill()`.

Task Priority Levels (`TaskPriority`)

Hydra supports three distinct priority strata:

1. `High` **(Priority 2)**: Time-critical device drivers and real-time event handlers.
2. `Normal` **(Priority 1)**: Standard userland processes, shell sessions, and background workers.

3. `Idle` **(Priority 0)**: Lowest priority; assigned exclusively to Task 1 (`idle`). It is only scheduled when no other `Ready` thread exists in the system.

Task Control Block Layout

```
pub struct Task {
    pub id: TaskId,
    pub name: &'static str,
    pub state: TaskState,
    pub priority: TaskPriority,
    pub regs: TaskRegisters,
    pub stack: Option<Box<[u8]>>,
    pub ticks_run: u64,
    pub timeslice_remaining: u32,
    pub cspace: Box<CapabilityTable>,
}
```

Memory Footprint Optimization & Stack Safety

- **Heap-Allocated CSpace**: The `CapabilityTable` provides 256 capability slots (~8 KiB). In early designs, allocating this structure inline inside `Task` by value resulted in stack overflows during task creation. By storing `cspace: Box<CapabilityTable>` , the memory footprint of `Task` on the stack is reduced from **8,256 bytes to ~168 bytes**, guaranteeing zero stack smashing during deep call chains.
- **Dedicated 16 KiB Kernel Stacks**: Newly spawned threads allocate `16,384 bytes` (`16 KiB`) on the 8 MiB kernel heap. The top of the stack is 16-byte aligned (`(stack_top & !0xF)`), and an initial 48-byte shadow space offset is reserved to satisfy Microsoft x64 / UEFI and SysV AMD64 calling convention requirements.

STAGE 9: MULTITASKING & SCHEDULER • BOOT SEQUENCE ORDER #29

Preemptive Multitasking & Context Switching

Overview

The Hydra multitasking subsystem implements a **preemptive round-robin scheduler** driven by the 8254 Programmable Interval Timer (PIT) firing at 100 Hz on IDT Vector 32 (IRQ 0). Multitasking enables concurrent execution of independent kernel threads, background worker tasks, and the interactive shell with transparent hardware context switching.

The implementation resides in `src/task/scheduler.rs`, with the public API exposed through `src/task/mod.rs` and interactive shell commands in `src/shell/commands/task_cmds.rs`.

Scheduling Architecture & Timeslicing

The scheduler operates on a fixed timeslice quantum of **5 PIT timer ticks (50 ms)** at 100 Hz.

```
sequenceDiagram
    autonumber
    participant Hardware as 8254 PIT Timer
    participant IDT as IDT Vector 32 (IRQ 0)
    participant Sched as Preemptive Scheduler
    participant CPU as x86_64 CPU Core

    Hardware->>IDT: Timer Interrupt (every 10 ms)
    IDT->>IDT: common_trap_handler: Push GPRs & Error Code
    IDT->>Sched: pit::on_timer_tick() & pic::send_eoi(0)
    IDT->>Sched: SCHEDULER.lock().schedule(&mut ExceptionContext)
    alt Timeslice > 1 Tick Remaining
        Sched->>CPU: timeslice_remaining -= 1 (No switch)
    else Timeslice Expired (0 Ticks)
        Sched->>Sched: Save active task registers from ExceptionContext
        Sched->>Sched: Wake up sleeping tasks (ticks ≥ wake_tick)
        Sched->>Sched: Select next Ready task (Round-Robin)
        Sched->>Sched: Restore next task registers into ExceptionContext
        Sched->>IDT: Return to assembly trampoline
        IDT->>CPU: Pop GPRs & iretq (Resumes new task context)
    end
```

Scheduling Algorithm Details

1. **Sleep Accounting**: On every tick, the scheduler scans all registered tasks and transitions any task in `TaskState::Sleeping { wake_tick }` back to `TaskState::Ready` if the current PIT tick count has reached or exceeded `wake_tick`.
2. **Timeslice Depletion**: The currently running task has its `ticks_run` counter incremented and `timeslice_remaining` decremented. If more than 1 tick remains in its timeslice, the scheduler returns immediately without context switching.
3. **Round-Robin Selection**: When the timeslice expires, the outgoing task's registers and `CR3` paging root are saved from `ExceptionContext`. The run-queue is scanned in two passes: - **Pass 1 (Normal & High Priority)**:

Scans for the next runnable non-idle task (`priority > Idle`). - **Pass 2 (Idle Fallback)**: If no normal tasks are ready, schedules Task 1 (`idle`) to execute low-power CPU halts (`hlt`).

4. **Context Switch Restoration**: The chosen task's registers are written back into `ExceptionContext` , its timeslice is reset to 50 ms, and if the task has a distinct `CR3` page table, the CPU paging root is updated via `Cr3::write()` .

Hardware Context Switching via `iretq`

Hydra employs an in-place stack frame modification pattern for architectural context switching:

1. When IRQ 0 fires, the CPU pushes `SS` , `RSP` , `RFLAGS` , `CS` , and `RIP` onto the active stack.
2. The assembly entry point `common_trap_handler` pushes all fifteen general-purpose registers (`RAX` through `R15`) and points `RSP` to the `ExceptionContext` .
3. `SCHEDULER.lock().schedule(ctx)` modifies the values stored inside the `ExceptionContext` on the stack.
4. As `common_trap_handler` unwinds, `pop r15..rax` loads the incoming task's general-purpose registers.
5. The hardware `iretq` instruction pops `RIP` , `CS` , `RFLAGS` , `RSP` , and `SS` . In 64-bit long mode, `iretq` unconditionally pops `RSP` and `SS` regardless of privilege level, atomically switching execution to the new task's stack and instruction pointer.

Task Trampoline & Calling Conventions

When a newly spawned kernel thread starts execution for the first time, it does not resume from a previously interrupted state. Instead, it enters through the **Task Trampoline**:

```
extern "C" fn task_trampoline(entry_fn_addr: usize) -> ! {
    let func: fn() = unsafe { core::mem::transmute(entry_fn_addr) };
    func();

    // Mark task Dead upon normal function exit and yield CPU
    crate::task::exit();
}
```

ABI Alignment & Register Conventions

- **Win64 vs SysV Dual Argument Passing**: Under the UEFI target (`x86_64-unknown-uefi`), the Microsoft x64 calling convention passes the first function argument in `RCX` (in contrast to SysV AMD64, which uses `RDI`). `spawn_internal` initializes both `regs.rcx` and `regs.rdi` to the entry point address to guarantee cross-ABI compatibility.
- **Shadow Space & 16-Byte Stack Alignment**: The initial stack pointer is configured as `initial_rsp = stack_top.saturating_sub(48)` . This satisfies the mandatory 32-byte Microsoft x64 shadow spill space requirement and aligns the stack to a 16-byte boundary upon entry into `func()` .

Background Idle Thread (`idle`)

Task 1 is initialized as the background `idle` thread with `TaskPriority::Idle` (Priority 0):

```
fn idle_task_entry() {
    loop {
        if x86_64::instructions::interrupts::are_enabled() {
            x86_64::instructions::hlt();
        } else {
            core::hint::spin_loop();
        }
    }
}
```

When no worker threads or shell tasks are runnable, the CPU enters the low-power halt state (`hlt`). The next timer tick or PS/2 keyboard interrupt automatically wakes the core to evaluate task readiness.

Public Multitasking API

The `src/task/mod.rs` module provides a unified public interface for thread management:

Function	Signature	Description
`init`	`fn init(current_cr3: u64)`	Initializes scheduler, creates Task 0 (`shell`), and spawns Task 1 (`idle`).
`is_active`	`fn is_active() -> bool`	Returns true if the multitasking subsystem is active.
`spawn`	`fn spawn(name, entry_fn, priority) -> TaskId`	Spawns a new concurrent kernel thread with a 16 KiB stack.
`yield_now`	`fn yield_now()`	Cooperatively relinquishes the remaining timeslice to the next task.
`sleep_ms`	`fn sleep_ms(ms: u64)`	Suspends the calling task for a duration in milliseconds.
`sleep_ticks`	`fn sleep_ticks(ticks: u64)`	Suspends the calling task for a specific number of 10ms PIT ticks.
`exit`	`fn exit() -> !`	Marks the calling task as `Dead` and yields the CPU permanently.
`kill`	`fn kill(id: TaskId) -> bool`	Terminates an active background thread by its Task ID.
`get_task_snapshots`	`fn get_task_snapshots() -> Vec<TaskSnapshot>`	Captures diagnostic thread telemetry for shell commands.

Shell Commands for Task Inspection & Control

Three built-in commands provide runtime task management in the interactive shell:

1. `ps` / `tasks` / `threads`

Displays all registered tasks, thread states, priorities, CPU tick consumption, and stack pointers:

```
hydra:/bin> ps
PID   NAME             STATE        PRIORITY        TICKS        STACK (RSP)
──────────────────────────────────────────────────────────────────────────────
0     shell            Running      Normal              0 0x0000000000000000
1     idle             Ready        Idle                0 0x00000000017a3150
2     bg_worker        Ready        Normal              0 0x00000000017a8950
──────────────────────────────────────────────────────────────────────────────
Total tasks: 3 | Timeslice: 50 ms | Worker counter: 0
```

2. `spawn [worker]`

Spawns a new concurrent background kernel thread executing a counter loop with `task::sleep_ms(100)`.

```
hydra:/bin> spawn worker
[+] Spawned background worker task #2 ('bg_worker').
    Type 'ps' to observe the task executing concurrently.
```

3. `kill <pid>`

Terminates an active background task by its numeric PID (system tasks PID 0 `shell` and PID 1 `idle` are protected from termination):

```
hydra:/bin> kill 2
[+] Task #2 terminated.
```

STAGE 9: MULTITASKING & SCHEDULER • BOOT SEQUENCE ORDER #30

Microkernel Synchronization Primitives & Wait Queues

Overview

In **Phase 4**, Hydra OS activates the `TaskState::Blocked` lifecycle state in the scheduler and introduces true sleeping synchronization primitives. Unlike busy-wait spinlocks that burn CPU cycles, sleep-locks transition contended threads to a suspended state and deschedule them immediately. When the lock or resource becomes available, the wait-queue awakens the thread, returning it to `TaskState::Ready`.

The synchronization subsystem is implemented in `src/sync/` and integrated with the preemptive scheduler in `src/task/scheduler.rs` and the IPC subsystem in `src/ipc/endpoint.rs`.

Core Primitives

1. Wait Queue (`WaitQueue`)

The `WaitQueue` maintains a FIFO queue of suspended tasks (`TaskId`) awaiting an event:

```
pub struct WaitQueue {
    waiters: VecDeque<TaskId>,
}
```

- `wait(&mut self)`: Captures the current `TaskId`, enqueues it, marks the task as `TaskState::Blocked`, and invokes `yield_now()` to relinquish the CPU immediately.
- `wake_one(&mut self) → Option<TaskId>`: Dequeues the oldest waiting task and transitions it to `TaskState::Ready`.
- `wake_all(&mut self) → usize`: Dequeues and unblocks all waiting tasks.

---

2. Sleeping Mutex (`Mutex<T>`)

The `Mutex<T>` provides mutual exclusion with automatic thread suspension:

```
pub struct Mutex<T> {
    locked: AtomicBool,
    owner: AtomicU64,
    waiters: SpinLock<WaitQueue>,
    data: UnsafeCell<T>,
}
```

- **Fast Path**: Uncontended acquisition succeeds via atomic CAS (`compare_exchange(false, true)`), requiring zero context switches or allocations.
- **Contention Path**: If already locked, the calling task registers in `waiters`, transitions to `TaskState::Blocked`, and yields the CPU.

- **Release**: When the `MutexGuard` is dropped, ownership is cleared, and `waiters.wake_one()` is called to reschedule the next waiter.

```
sequenceDiagram
    autonumber
    actor ThreadA as Thread A
    actor ThreadB as Thread B
    participant Lock as Mutex<T>
    participant Sched as Scheduler

    ThreadA->>Lock: lock() (CAS false -> true)
    Note over Lock: Fast path acquired! Owner=Thread A
    ThreadB->>Lock: lock() (contended)
    Note over Lock: Enqueues Thread B into WaitQueue
    Lock->>Sched: block_current() (TaskState::Blocked)
    Note over Sched: Thread B descheduled
    ThreadA->>Lock: unlock() (MutexGuard drop)
    Lock->>Sched: unblock_task(Thread B)
    Note over Sched: Thread B moved to TaskState::Ready
    Sched->>ThreadB: Resumes execution, acquires lock
```

---

3. Counting Semaphore (`Semaphore`)

The `Semaphore` manages shared permits: - **`acquire()`**: Atomically decrements `permits`. If no permits remain, the caller is enqueued in the semaphore's `WaitQueue` and suspended. - **`release()`**: Increments `permits` and awakens one waiting task from the queue.

---

4. Condition Variable (`CondVar`)

The `CondVar` enables threads to suspend execution until a condition predicate is satisfied: - **`wait(guard: MutexGuard<T>)`**: Atomically releases the held mutex, enqueues the thread in the condition variable's wait queue, and blocks. When awakened, it re-acquires the mutex before returning. - **`notify_one()`**: Wakes a single waiting thread. - **`notify_all()`**: Wakes all waiting threads.

---

5. Synchronized Bounded Channel (`Channel<T>`)

The `Channel<T>` implements thread-safe message passing: - **`send(&self, item: T)`**: Blocks if the channel buffer is at capacity until a receiver consumes data. - **`recv(&self) -> T`**: Blocks if the channel is empty until a sender transmits an item.

Blocking Inter-Process Communication (IPC)

In `src/ipc/endpoint.rs`, microkernel endpoints feature integrated send and receive wait queues:

```
pub struct Endpoint {
    id: u32,
    buffer: [Option<Message>; QUEUE_CAPACITY],
    head: usize,
    tail: usize,
    count: usize,
    send_waiters: WaitQueue,
    recv_waiters: WaitQueue,
}
```

- `send_blocking(msg)`: Transmits a message, suspending the sender if the queue is full. Awakens any blocked receivers.
- `receive_blocking() → Message`: Extracts a message, suspending the receiver if the queue is empty. Awakens any blocked senders.
- **System Endpoints (`SYSTEM_ENDPOINTS[0..15]`)**: Global shared rendezvous ports accessible both by kernel threads and user processes via `SYS_IPC_SEND` and `SYS_IPC_RECV` system calls.

Interactive Shell Verification

The subsystem is fully verified in the shell:

```
hydra:/bin> sync

============================================================
[+] HYDRA MICROKERNEL: SYNCHRONIZATION & WAIT QUEUES
============================================================
  Sleep-Lock Mutex Status (DEMO_MUTEX):
    - State           : UNLOCKED
    - Owner PID       : 0
    - Blocked Waiters : 0

  Active System IPC Endpoints (0..15):
    EP_ID PENDING_MSGS WAIT_RECEIVERS WAIT_SENDERS
    ------------------------------------------------
    0     0            0              0
    1     0            0              0
    2     0            0              0

  Scheduler Task State Summary:
    - Running: 1 | Ready: 1 | Blocked: 0 | Sleeping: 0
============================================================

hydra:/bin> spawn consumer
[+] Spawned consumer task #4 waiting on Endpoint #1.
    Type 'ps' to observe consumer transitioning into Blocked state.
    Run 'spawn producer' to deliver messages and unblock it.
hydra:/bin>  [Consumer] Calling receive_blocking() on Endpoint #1 (will Block if empty)...
ps
PID   NAME        MODE    STATE     PRIORITY    TICKS        STACK (RSP)
-------------------------------------------------------------------------
0     shell       Kernel  Running   Normal        749 0x0000000001fe5e4e0
1     idle        Kernel  Ready     Idle            0 0x00000000017a3190
2     worker_a    Kernel  Dead      Normal          2 0x00000000017a8938
3     worker_b    Kernel  Dead      Normal          3 0x00000000017ae148
4     consumer    Kernel  Blocked   Normal          1 0x00000000017b3828
-------------------------------------------------------------------------

hydra:/bin> spawn producer
[+] Spawned Producer task #5 feeding Endpoint #1.
hydra:/bin>  [Producer] Depositing message #1 to Endpoint #1...
[Producer] Message #1 sent.
[Consumer] Received message from Endpoint #1: label=0x1001, arg0=1
[Producer] Depositing message #2 to Endpoint #1...
[Producer] Message #2 sent.
[Consumer] Received message from Endpoint #1: label=0x1002, arg0=2
[Producer] Depositing message #3 to Endpoint #1...
[Producer] Message #3 sent.
[Producer] Completed sending batch.
[Consumer] Received message from Endpoint #1: label=0x1003, arg0=3
[Consumer] Completed receiving batch.
```

STAGE 10: INTERACTIVE SHELL • BOOT SEQUENCE ORDER #31

Interactive Microkernel Shell Documentation

- **Source Directory**: `src/shell/`
- **Subsystem**: Microkernel User Interface & Command REPL

---

1. Conceptual Overview

Operating system development often starts with non-interactive boot routines that print diagnostic logs and halt. To enable active exploration, debugging, and experimentation with kernel primitives on live hardware or virtual machines, Hydra provides an **Interactive Microkernel Shell**.

The shell is architected as a modular, extensible command registry where every command or tool lives in its own dedicated source file:

```
src/shell/
├── mod.rs                  # REPL loop, banner, prompt formatting, and command dispatcher
├── state.rs                # ShellState (CSpace, IPC endpoint, prompt color, cwd, history)
├── command.rs              # The `Command` trait and argument parsing helpers
├── editor.rs               # Interactive line editor (history cycling, tab completion, arrow keys)
├── history.rs              # Command history ring buffer (32 entries)
└── commands/               # Modular Command Implementations
    ├── mod.rs              # Static command registry (`COMMANDS`)
    ├── gnu_fs.rs           # GNU Filesystem utils (`pwd`, `cd`, `mkdir`, `rmdir`, `rm`, `cp`, `mv`, `
    ├── gnu_text.rs         # GNU Text inspection (`head`, `tail`, `wc`, `grep`, `basename`, `dirname`
    ├── gnu_sys.rs          # GNU System & scripting (`uname`, `whoami`, `sleep`, `seq`, `yes`, `true`
    ├── fs_cmds.rs          # File operations (`ls`, `cat`, `write`)
    ├── history_cmd.rs      # Session history viewer (`history`)
    ├── script.rs           # Bash / POSIX script runner (`sh`, `source`, `.`)
    ├── help.rs             # Help command & usage manual generator
    ├── info.rs             # OS version & architecture overview (`info`, `version`)
    ├── status.rs           # Subsystem health inspection
    ├── cpu.rs              # CPU control registers (`cpu`)
    ├── mem.rs              # Memory statistics & allocation (`mem`, `alloc`)
    ├── translate.rs        # Virtual-to-physical paging translation (`translate`)
    ├── caps.rs             # CSpace inspection & revocation (`caps`, `cap_insert`, `cap_revoke`)
    ├── ipc.rs              # Microkernel message rendezvous (`ipc`)
    ├── time.rs             # Real-Time Clock (`time`, `date`)
    ├── power.rs            # ACPI/UEFI power management (`shutdown`, `reboot`, `halt`)
    ├── zfs_cmds.rs         # OpenZFS administration (`zpool`, `zfs`)
    ├── efifs.rs            # Low-level UEFI FAT boot volume diagnostics (`efifs`)
    └── misc.rs             # Utilities (`color`, `echo`, `clear`, `panic`)
```

Zsh-Style Interactive Line Editor & History

The shell includes an interactive line editor (`editor.rs`): - **Up / Down Arrow History Navigation**: Traverses the command history buffer (`history.rs`). Pressing ↑ recalls previously entered commands, while ↓ cycles forward or restores the typed draft line. - **Tab Auto-Completion (Zsh Style)**: Pressing `Tab` auto-completes built-in command names, aliases, and filesystem paths in the current directory. When multiple matches exist, a color-coded column listing is displayed, and the prompt and current editing line are cleanly reprinted. - **In-Line Cursor Navigation**: ← and → move the cursor anywhere within the line; `Home` jumps to the start, `End` jumps to the end. - **In-Line Editing**: Character

insertion, `Backspace`, and `Delete` work anywhere in the line with automatic redraw. - **History Expansion** `!!`: Typing `!!` echoes and immediately re-executes the most recent command.

Standard Terminal Key Combinations

The interactive line editor and execution runtime implement standard POSIX / Readline / Zsh control key bindings:

Key Combination	Action / Function	Behavior
`Ctrl+C`	**Interrupt / Cancel**	While editing: cancels current input, prints `^C`, and restarts prompt on a new line. While running a program (`yes`, `sleep`, `seq`, `sh`, `cat`, `grep`, `head`, `tail`): halts execution immediately with status `130`.
`Ctrl+D`	**EOF / Forward Delete**	If the line buffer is empty: signals EOF (`logout` / clean shutdown). If the line is non-empty: deletes character under cursor (same as `Delete`).
`Ctrl+L`	**Clear Screen**	Clears the UEFI display and redraws the shell prompt and current line buffer at row 0.
`Ctrl+A`	**Line Start**	Jumps cursor to the beginning of the line (equivalent to `Home`).
`Ctrl+E`	**Line End**	Jumps cursor to the end of the line (equivalent to `End`).
`Ctrl+U`	**Kill Line Before**	Deletes all text from the start of the line up to the cursor.
`Ctrl+K`	**Kill Line After**	Deletes all text from the cursor to the end of the line.
`Ctrl+W`	**Kill Word Backward**	Erases the previous whitespace-delimited word before the cursor.
`Ctrl+P`	**Previous History**	Cycles backward in command history (equivalent to ↑).
`Ctrl+N`	**Next History**	Cycles forward in command history (equivalent to ↓).
`Ctrl+B`	**Move Backward**	Moves cursor left by one character (equivalent to ←).
`Ctrl+F`	**Move Forward**	Moves cursor right by one character (equivalent to →).
`Ctrl+H`	**Backspace**	Deletes character to the left of the cursor.
`Ctrl+Z`	**Suspend**	Displays `^Z` notification.

Shell Script Runner (Bash Compatible)

The shell provides a built-in script interpreter (`sh`, `source`, `.`) capable of executing `.sh` files stored in ZFS or EFI boot volumes: - Supports `#` comment lines and shebangs (`#!/bin/sh`, `#!/bin/bash`). - Supports command chaining with `;` (e.g. `mkdir /tmp/test; cd /tmp/test; touch a.txt; ls`). - Supports script arguments (`$1`, `$2`, `$0`) and last exit status (`$?`). - Pre-loaded with `/bin/demo.sh` demonstrating automated multi-command execution.

Terminal Scrolling via Scroll Lock & Arrows

Hydra implements classic PC terminal scrolling using the **Scroll Lock** mode and arrow keys:

1. **When Scroll Lock is Active (`[SCROLL]` Prompt)**: - The shell freezes command execution and converts navigation keys into terminal viewport navigation:
    - ↑ (Up Arrow): Scrolls the terminal screen **UP** line-by-line into the 512-line kernel scrollback buffer.
    - ↓ (Down Arrow): Scrolls the terminal screen **DOWN** line-by-line toward the active command prompt.
    - Page Up: Scrolls up by an entire screen page (`rows - 2` lines).
    - Page Down: Scrolls down by an entire screen page.
    - **Full Color & Attribute Preservation**: The scrollback buffer tracks both character bytes and formatting attributes (`(fg & 0xF) | ((bg & 0x7) << 4)`). All syntax highlighting, prompt colors (`LightCyan`), section headers (`Yellow`), directory names (`Green`), and error messages (`LightRed`) are preserved verbatim when scrolling up and down.
    - When scrolled up into history, a cyan status banner displays the visible line range: `-- SCROLL LOCK (up N lines, lines X-Y of Z) [UP/DOWN or PgUp/PgDn, ESC to exit] --`.
    - Allows naturally reviewing long command outputs (like `help`, `cat`, `ls -l`, `history`, or kernel boot diagnostics) that scrolled off the top of the screen without losing any visual hierarchy.
2. **When Scroll Lock is Inactive (Normal Mode)**: - ↑ (Up Arrow) cycles backward through the command history ring buffer. - ↓ (Down Arrow) cycles forward through the command history ring buffer. - Typing any printable character, pressing Enter, or hitting Backspace automatically snaps the viewport back to the active prompt.

3. **Activating & Toggling Scroll Lock**: - **Keyboard Shortcut**: Press Ctrl+S or F12 in the shell to immediately toggle Scroll Lock on/off. The prompt dynamically reflects the mode (`hydra:/bin [SCROLL]>` vs `hydra:/bin>`). - **Direct Page Up**: Pressing Page Up at the prompt automatically enables scrolling mode and pages up through previous screen contents. - **Shell Command**: Run `scroll` (or `scroll on` / `scroll off`) from the command line. - **Exiting Scroll Mode**: Press Escape, Ctrl+S, or Ctrl+C to return to normal prompt mode. - **VirtualBox Note**: In VirtualBox with UEFI enabled, the firmware's `EFI_SIMPLE_TEXT_INPUT_PROTOCOL` does not generate key events for the physical Scroll Lock key (and host OS keyboard hooks often intercept or discard it). Use Ctrl+S, F12, or `scroll on` inside the VM for instant, reliable scrolling.

Dynamic Working Directory Prompt

The shell tracks the current working directory dynamically in `ShellState` (initialized to `/bin`), displaying a standard Unix prompt:

```
hydra:/bin> pwd
/bin
hydra:/bin> cd /
hydra:/> ls
NAME                    TYPE         CHECKSUM     STATUS
-------------------------------------------------------------
bin                     <DIR>        Fletcher4     ONLINE
root                    <DIR>        Fletcher4     ONLINE
tmp                     <DIR>        Fletcher4     ONLINE
-------------------------------------------------------------
hydra:/> cd /bin
hydra:/bin>
```

---

2. Command Reference

GNU Core Utilities (POSIX / Linux Compatible)

Command	Syntax	Description	Supported Options & Flags
`pwd`	`pwd`	Prints current working directory.	
`cd`, `chdir`	`cd [dir]`	Changes current working directory. Default: `/root`.	Relative paths, `..`, `.`, `/`
`ls`, `dir`	`ls [path]`	Lists files and directories in ZFS. Default: `cwd`.	`-l` (long format), `-a` (show all)
`cat`, `type`	`cat <file>`	Displays file contents verified with Fletcher-4.	Path resolution relative to `cwd`
`more`, `less`	`more <file>`	Paginates file with interactive Up/Down and PageUp/PageDown scrolling.	`less`, `page` aliases
`write`	`write <file> [text]`	Writes/appends data to a file via CoW transaction.	Path resolution relative to `cwd`
`touch`	`touch <file ... >`	Creates empty file or updates birth TXG / timestamp.	Path resolution relative to `cwd`
`mkdir`	`mkdir <dir ... >`	Creates one or more directories in ZFS.	`-p` (create parents without error)
`rmdir`	`rmdir <dir ... >`	Removes empty directories from ZFS.	Verifies directory empty count
`rm`, `unlink`	`rm <file ... >`	Removes files or directories from ZFS.	`-r` / `-R` (recursive), `-f` (force)
`cp`, `copy`	`cp <src> <dst>`	Copies files within ZFS pool via CoW.	Directory destination resolution
`mv`, `rename`	`mv <src> <dst>`	Moves or renames files or directories in ZFS.	Cross-directory and in-place moves
`stat`	`stat <file>`	Displays detailed file or directory metadata.	Size, blocks, birth TXG, checksum
`df`	`df`	Reports filesystem and storage pool capacity.	`-h` (human-readable: MiB/KiB)
`du`	`du [path]`	Estimates file and directory disk space usage.	`-h` (human-readable), `-s` (summary), `-a`
`head`	`head <file ... >`	Displays first N lines of files. Default: 10 lines.	`-n <num>` or `-<num>`
`tail`	`tail <file ... >`	Displays last N lines of files. Default: 10 lines.	`-n <num>` or `-<num>`

Command	Syntax	Description	Supported Options & Flags
`wc`	`wc <file ... >`	Counts newlines, words, and bytes in files.	`-l` (lines), `-w` (words), `-c` (bytes)
`grep`	`grep <pattern> <file>`	Searches for patterns in files.	`-i` (ignore case), `-n` (line numbers), `-v` (invert), `-c` (count)
`basename`	`basename <path> [suf]`	Strips directory and suffix from filenames.	Optional suffix removal
`dirname`	`dirname <path>`	Strips last component from filename path.	
`uname`	`uname`	Prints operating system and architecture info.	`-a` (all), `-s` (kernel), `-n` (node), `-r` (release), `-v` (version), `-m` (arch)
`whoami`, `id`	`whoami`	Prints current user identity and capabilities.	`-u`, `-g`, `-a` (full capability context)
`sleep`	`sleep <secs>`	Delays execution for specified duration (up to 60s).	Non-blocking firmware stall
`seq`	`seq [FIRST [INC]] LAST`	Prints sequences of integers.	Customizable start, step, and end
`history`, `hist`	`history`	Displays numbered session command history list.	Up/down arrow key cycling
`sh`, `source`, `.`	`sh <script.sh>`	Executes Bash-compatible script files line by line.	`#` comments, `;` chaining, `$1..$9`, `$?`
`!!`	`!!`	Repeats and re-executes the last command.	Shell history expansion
`true`	`true`	Returns success status (exit status 0).	
`false`	`false`	Returns failure status (exit status 1).	

Microkernel & Hardware Diagnostics

Command	Syntax	Description
`help`, `?`	`help [cmd]`	Displays the interactive command manual and specific usage.
`info`, `version`	`info`	Prints Hydra Microkernel OS version, build target, and architecture overview.
`status`	`status`	Reports operational health across GDT, CPU, Frame Allocator, Paging, CSpace, and IPC.
`cpu`	`cpu`	Displays CPU control registers (`CR0`, `CR2`, `CR3` PML4 root, `CR4`).
`mem`, `memory`	`mem`	Displays physical memory manager statistics (Total Usable RAM, Allocated, Free).
`alloc`	`alloc`	Allocates a 4 KiB physical memory frame from the bump allocator and returns its physical address.
`translate`	`translate <hex_virt_addr>`	Walks the active 4-level PML4 page table hierarchy to resolve virtual to physical addresses.
`caps`, `cspace`	`caps`	Inspects the active task's capability table (slot numbers, resource types, and rights bitflags).
`cap_insert`	`cap_insert <ep_id>`	Inserts a new Endpoint capability into the task's CSpace.
`cap_revoke`	`cap_revoke <slot>`	Revokes an existing capability from the specified CSpace slot.
`ipc`	`ipc <label> [a0] [a1]`	Sends and receives a fast register IPC message through rendezvous Endpoint #1.
`time`, `date`	`time`	Queries the hardware Real-Time Clock (RTC) via UEFI runtime services.
`zpool`	`zpool <status|list>`	Inspects and manages ZFS storage pools (`hydrapool`, vdevs, CoW Merkle tree).
`zfs`	`zfs <list|snap|cat|write>`	Manages ZFS datasets (`hydrapool/root`, `hydrapool/bin`, `hydrapool/tmp`) and CoW snapshots.
`efifs`	`efifs <ls|cat|run>`	Low-level firmware diagnostic tool inspecting raw UEFI FAT bootloader media.
`color`	`color <name>`	Customizes shell prompt and text color (`cyan`, `green`, `yellow`, `red`, `white`, `magenta`, `blue`).
`scroll`	`scroll [on|off]`	Toggles or sets Scroll Lock mode for Up/Down arrow terminal scrolling.
`echo`	`echo <text ... >`	Echoes input arguments back to the screen.

Command	Syntax	Description
`clear`, `cls`	`clear`	Clears the active screen buffer and resets the cursor.
`panic`	`panic`	Manually triggers the kernel panic handler to verify diagnostic output.
`reboot`	`reboot`	Reboots the computer via 8042 keyboard pulse, PCI reset, and UEFI cold reset.
`shutdown`	`shutdown`	Powers down the computer via VirtualBox/QEMU ACPI I/O and UEFI shutdown.
`halt`	`halt`	Disables CPU interrupts and enters a low-power `hlt` loop.

---

3. Integrating New Commands & Standalone Programs

Hydra supports two complementary paradigms for adding commands and applications:

Approach A: Dynamic Filesystem Search ($PATH Discovery — No Code Changes!)

This is the recommended approach for integrating standalone applications, external utilities, and third-party binaries.

1. **How It Works**: When a command is entered (e.g. `hydra> matrix` or `hydra> test_app arg1 arg2`), the shell first checks built-in commands. If not found, it dynamically searches the `$PATH` directories on the boot volume: - `\bin\` - `\apps\` - `\EFI\BOOT\` - `\` (root directory)

For each directory, it probes for: - `<dir>\<cmd>.efi` - `<dir>\<cmd>`

2. **Loading & Execution**: - The file is located on the UEFI boot volume via `SimpleFileSystem`. - The binary's PE32+ header (`MZ` signature) is verified. - Memory pages are dynamically allocated and the executable is loaded via `uefi::boot::load_image`. - Arguments are passed through the UEFI `LoadedImage` protocol. - The process is executed via `uefi::boot::start_image`. - Upon termination, control returns immediately to the Hydra interactive shell prompt!

3. **Zero Kernel Modifications**: You can drop any compiled UEFI PE32+ `.efi` executable into `\bin\` or `\apps\` on your boot USB or disk image at any time. **You do NOT need to modify `commands/mod.rs` or recompile the kernel!**

Approach B: Static Built-in Command Integration

Used for core microkernel primitives that require direct access to kernel internals, `CapabilityTable`, or `ShellState`.

Step 1: Create a New Command File

Create a new file in `src/shell/commands/`, e.g. `src/shell/commands/my_tool.rs`:

```rust
use crate::console::set_color;
use uefi::proto::console::text::Color;
use crate::shell::command::Command;
use crate::shell::state::ShellState;

pub struct MyToolCommand;

impl Command for MyToolCommand {
    fn name(&self) -> &'static str {
        "mytool"
    }

    fn aliases(&self) -> &'static [&'static str] {
        &["mt"]
    }

    fn description(&self) -> &'static str {
        "Demonstrates custom command integration"
    }

    fn usage(&self) -> &'static str {
        "mytool <argument>"
    }

    fn execute(&self, args: &[&str], state: &mut ShellState) {
        set_color(Color::LightGreen, Color::Black);
        println!("[+] MyTool executed with args: {:?}", args);
    }
}
```

Step 2: Register in `src/shell/commands/mod.rs`

In `src/shell/commands/mod.rs`: 1. Add the module: `pub mod my_tool;` 2. Export the struct: `pub use my_tool::MyToolCommand;` 3. Add `&MyToolCommand` to the `COMMANDS` slice: `rust pub static COMMANDS: &[&dyn Command] = &[ ... &MyToolCommand, ];`

---

4. Architecture & Key Features

Automatic 3-Second Bare-Metal Transition Countdown

Upon shell startup, Hydra automatically initiates a 3-second countdown before transitioning to bare-metal hardware ownership (`exitboot`): - **Hands-Free Boot**: Users no longer need to manually type `exitboot` to unlock Ring 3 user mode, autonomous framebuffer rendering, and preemptive multitasking. - **Countdown Display**: An interactive timer counts down (`3 ... 2 ... 1 ... NOW`), after which `ExitBootServices` is called, hardware ownership is established, and the bare-metal prompt `hydra:/bin>` appears. - **User Control**: - Pressing `Enter` during the countdown immediately bypasses the remaining delay and transitions to bare metal instantly. - Pressing any other key (e.g. `Space` or `Esc`) cancels the countdown, keeping the shell in UEFI Boot Services mode.

Thread-Safe Interactive Input & Automatic Prompt Refresh

Interactive input is handled by `src/shell/editor.rs` (`read_line`): - **Automatic Prompt Appearance**: When a command finishes execution, the shell REPL loop immediately formats and renders the `hydra:/>` prompt on screen without requiring the user to press `Enter`. - **Foreground vs. Background Task Synchronization**: Commands like `spawn user` , `spawn rogue` , `spawn mutex` , and `spawn producer` execute in the foreground by default using `task::wait_task(id)` , returning CPU control and rendering the prompt precisely when task execution finishes. Passing `--bg` , `-b` , or `&` launches the task asynchronously in the background. - **Asynchronous Prompt Auto-Refresh**: If a concurrent background task (e.g. `consumer` receiving IPC messages) writes output to the console while

the shell is waiting for input, `get_key_or_refresh()` monitors the global console generation counter (`CONSOLE_OUTPUT_GENERATION`). Once background output completes, the shell automatically moves to a new line and re-renders the prompt (and preserves any partially typed input) without requiring user intervention. - Real-time backspace support erases characters from screen and buffer seamlessly via cursor positioning. - Enter (`\r` or `\n`) submits the command line.

Multi-Tier Power Management (`shutdown` & `reboot`)

Virtual machine firmware (such as VirtualBox's EFI and QEMU/EDK2) frequently stubs or hangs on generic UEFI runtime `ResetSystem(EfiResetShutdown)` prior to ACPI table activation. Hydra solves this by implementing direct hardware I/O power transitions: 1. **VirtualBox ACPI**: Writes sleep state S5 bits to PIIX3/4 ACPI `PM1a_CNT` (I/O Port `0x4004`). 2. **QEMU ACPI**: Writes to standard PIIX4 ACPI `PM1a_CNT` (I/O Port `0x604`) and debug exit (`0x501`). 3. **Cloud / Bochs ACPI**: Writes to I/O Port `0xb004` and APM (`0xb2`). 4. **Reboot Sequencing**: Pulses the 8042 keyboard controller reset line (`0x64`, `0xFE`) and the PCI reset control register (`0xCF9`, `0x06`) before falling back to UEFI cold reset.

Virtual Address Translation Inspection

The `translate` command allows live debugging of paging:

```
hydra> translate 0x400000
[*] Translating Virtual Address 0x400000 via PML4 tree: Mapped → Physical Address 0x400000
```

On-Demand Memory Allocation

The `alloc` command demonstrates the physical memory frame allocator in real-time:

```
hydra> alloc
[+] Successfully allocated 4 KiB frame at physical address: 0x101000
hydra> mem
Physical Memory Manager Statistics:
  Total Usable RAM:    ...
  Allocated Frames:    ...
```

Live Microkernel IPC

The `ipc` command sends a message through a rendezvous endpoint and reads the delivered payload:

```
hydra> ipc 0x2001 100 200 300
[+] Message queued to Endpoint #1: label=0x2001, args=[100, 200, 300, 0]
[+] Rendezvous complete! Received message from Endpoint #1:
    - Label: 0x2001
    - Args:  [100, 200, 300, 0]
```